UNA PANORAMICA NEUROFUNCIONAL DEL CEREBRO

UNIDAD DE NEUROPSICOLOGÍA CLÍNICA udp
FACULTAD DE PSICOLOGÍA

CLÍNICA PSICOLÓGICA udp
FACULTAD DE PSICOLOGÍA



OBJETIVOS

Introducir al estudiante en la relación de diversas estructuras cerebrales y un diverso rango de funciones cognitivas, conductuales y socio- emocionales





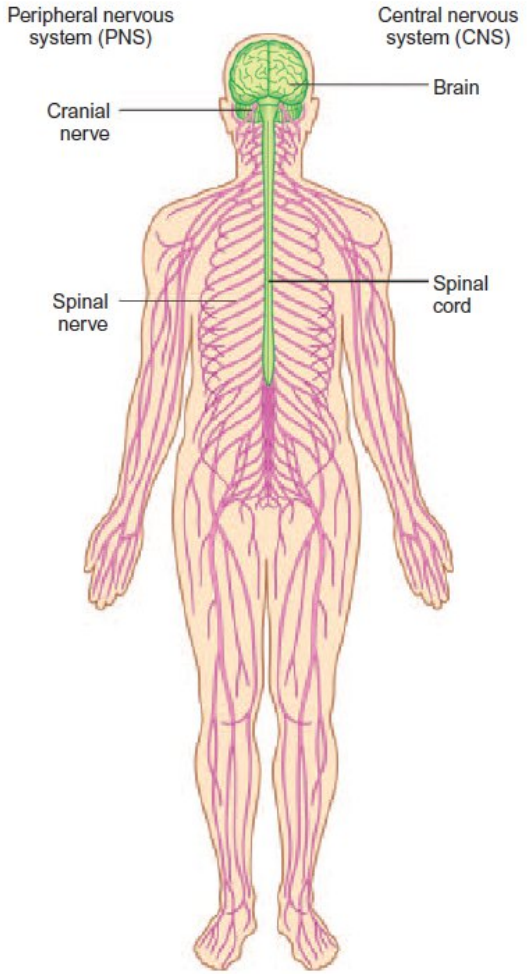


FIGURE 5.1 The central and peripheral nervous systems. *Source*: Standring, 2005.

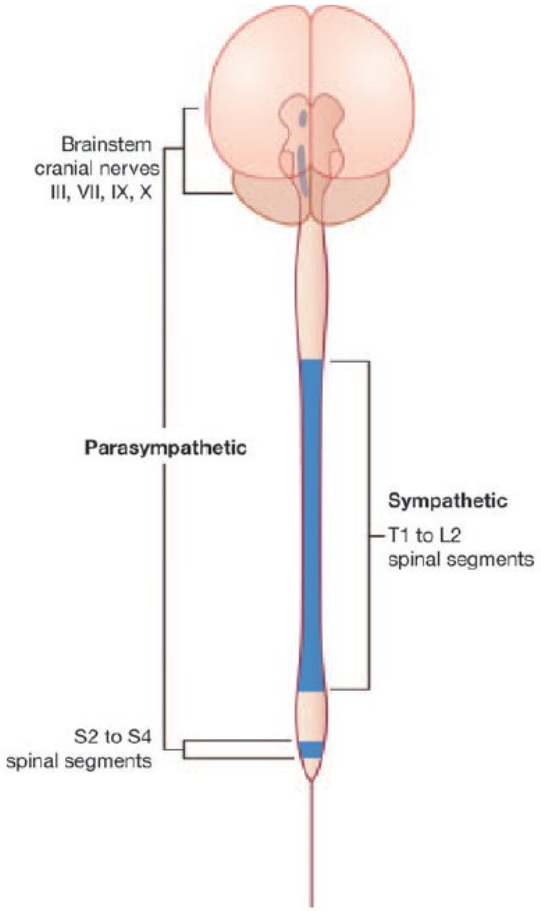
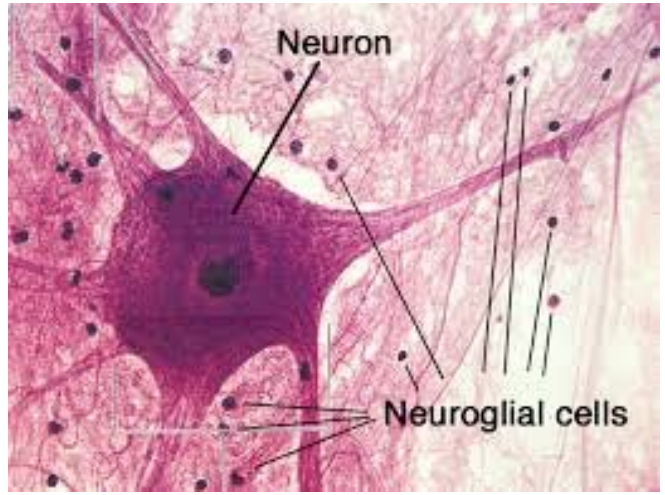
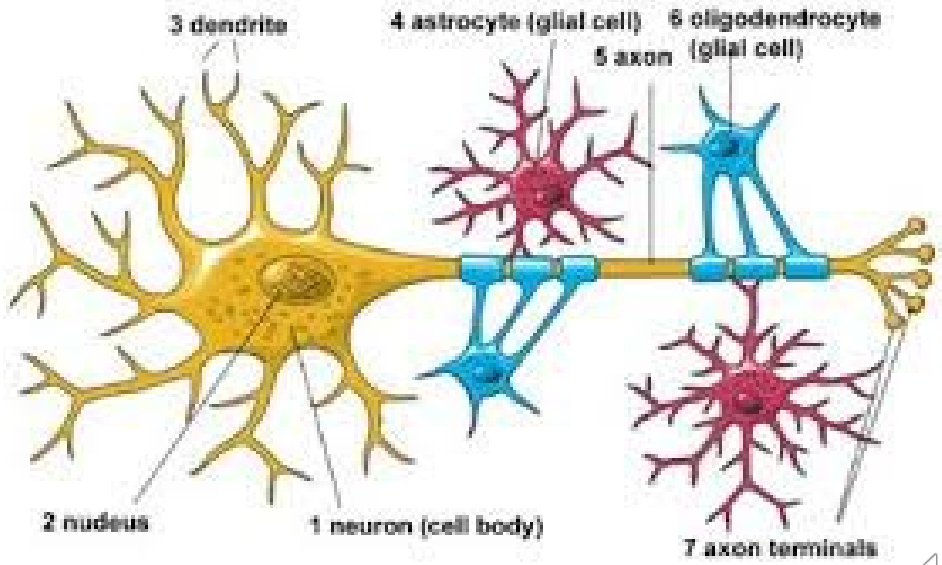
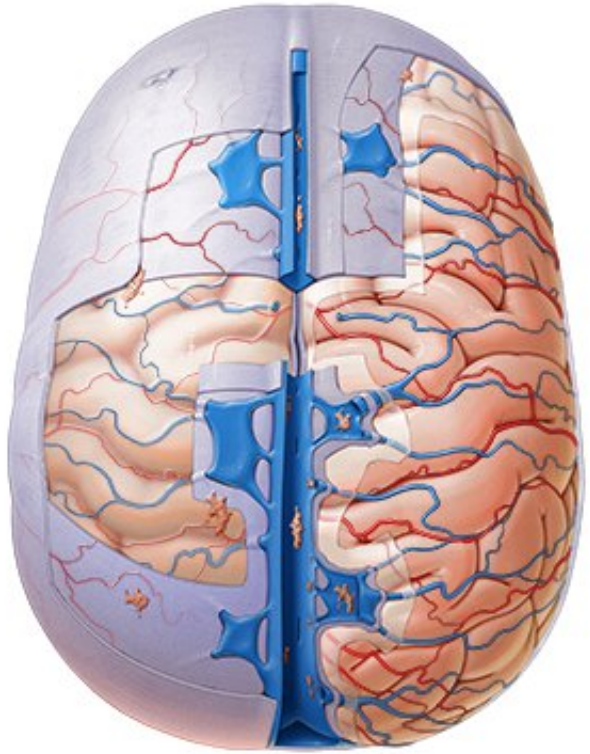
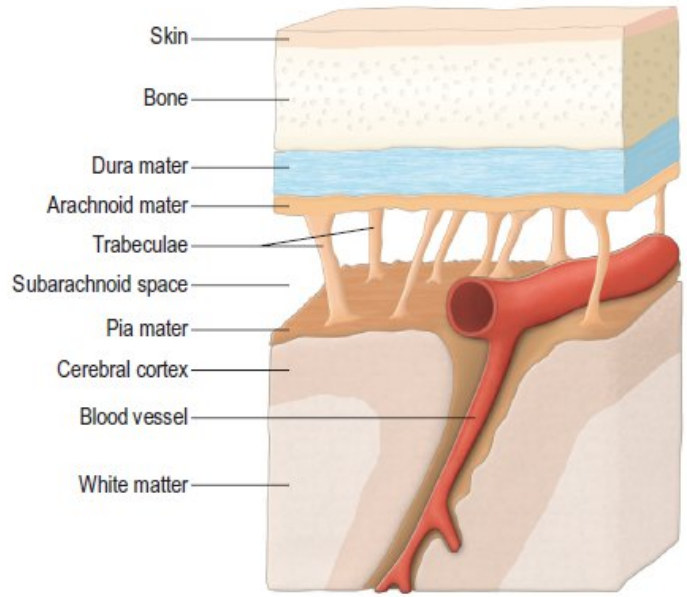


FIGURE 5.2 Parts of the central nervous system include the spinal cord and the brain. *Source*: Standring, 2005.









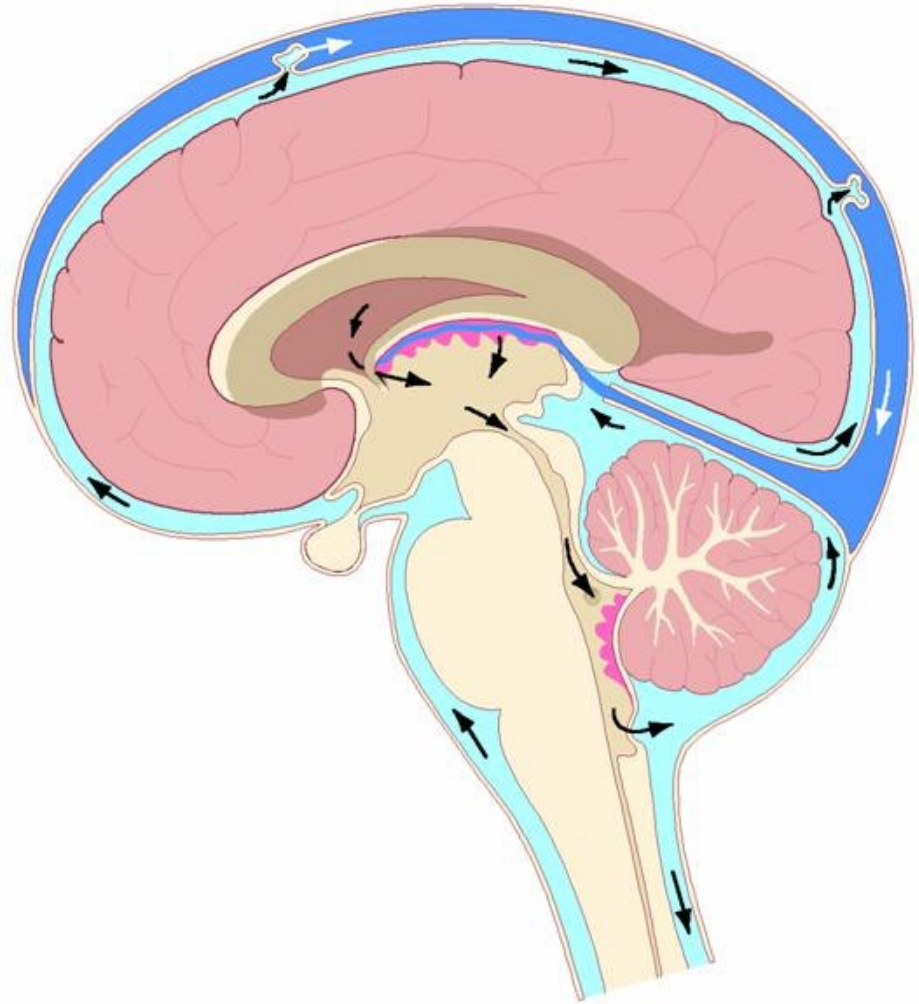


Figure 1.16 **A section through the skull, illustrating the relationships between the meninges and the CNS.**

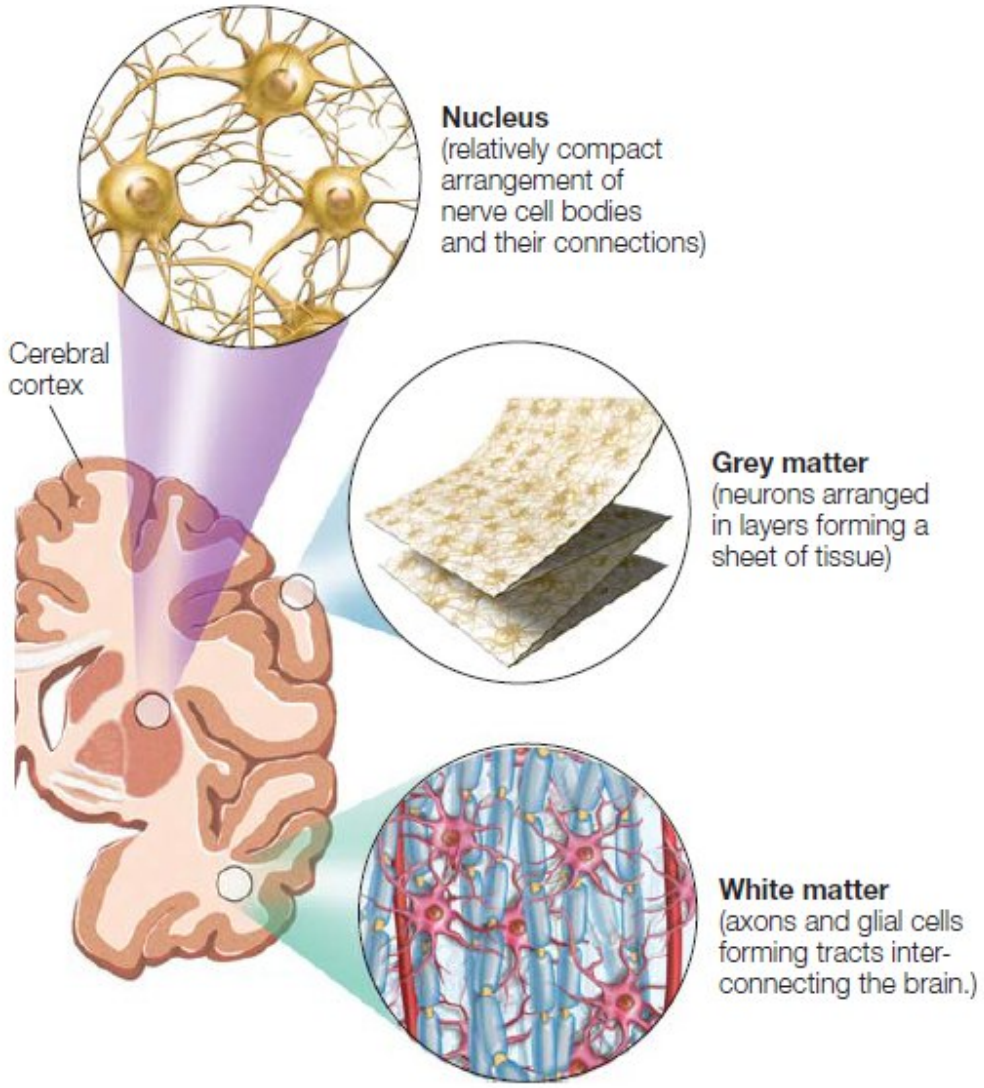


FIGURE 2.18 Organization of neurons in the CNS.
In the CNS, neurons can be organized in clumps called nuclei (top—not to be confused with the nucleus inside each neuron), which are most commonly found in subcortical and spinal structures, or sheets called layers (middle), which are most commonly found in the cortex. The cell bodies of glial cells are located in the white matter (e.g., oligodendrocytes), and in the cortex.

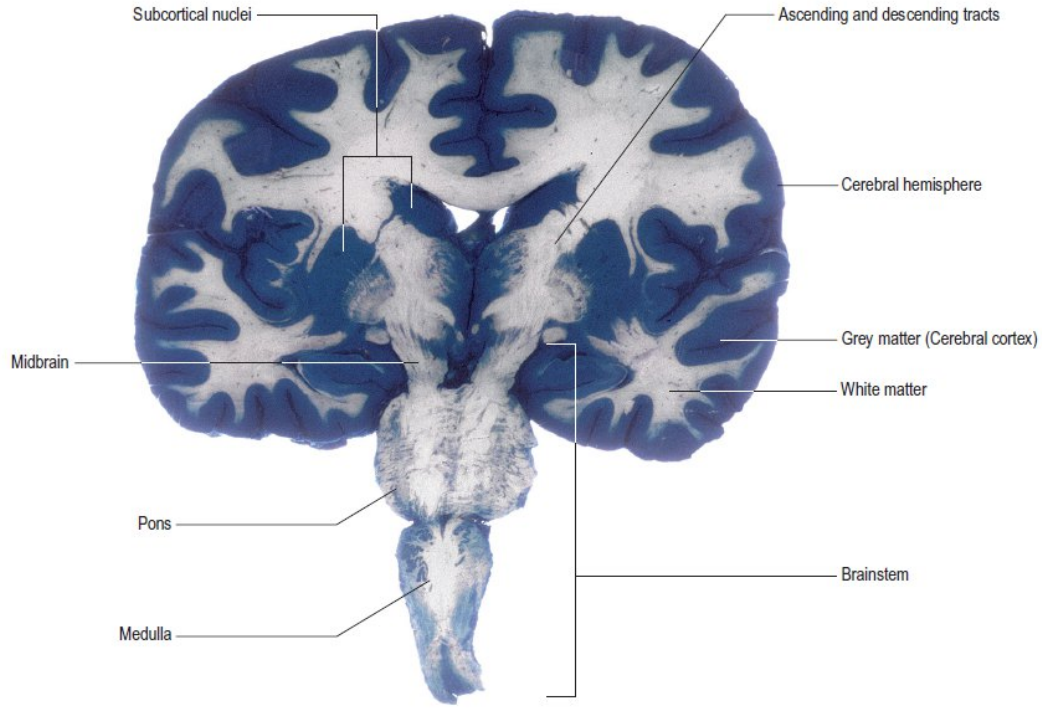


Figure 1.7 **Coronal section through the brain illustrating the distribution of grey and white matter.** The section has been stained by Mulligan's technique, which colours grey matter blue, leaving white matter relatively unstained.

MATERIA GRIS: CUERPOS CELULARES DE NEURONAS
MATERIAL BLANCA: AXONES Y CELULAS GLIALES

TRACTOS DE ASOCIACION NEURAL

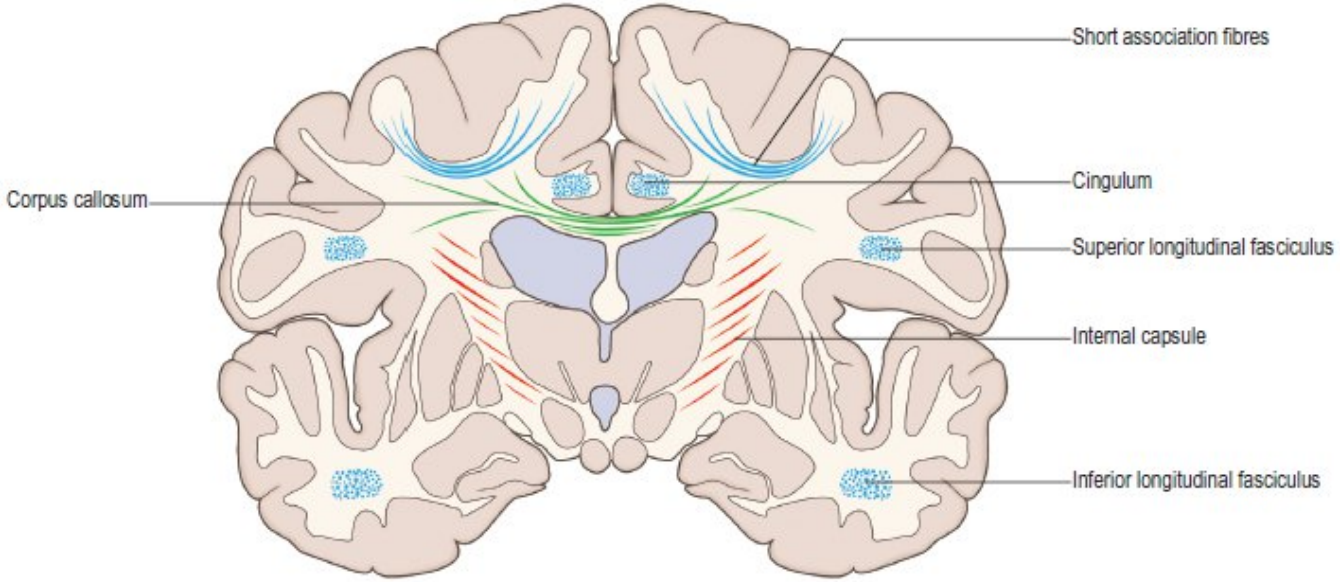


Figure 13.21 **Coronal section of the cerebral hemisphere.** The diagram shows the location of the principal association, commissural and projection fibres.

TRACTOS / COMISURAS

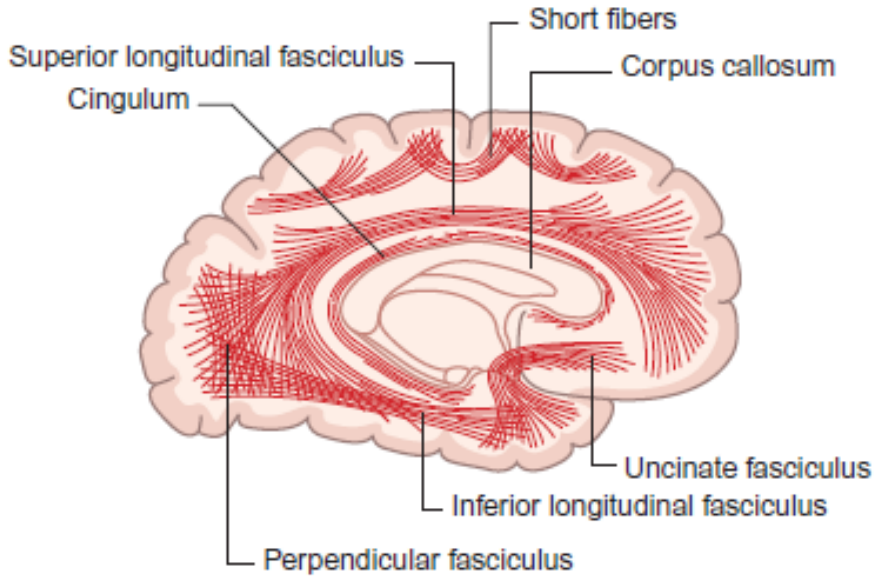


FIGURE 5.38 Schematic drawing of the connectivity of the brain, showing major fiber patterns. *Source*: Standring, 2005.

EVOLUCION Y ESTRUCTURA DEL CEREBRO

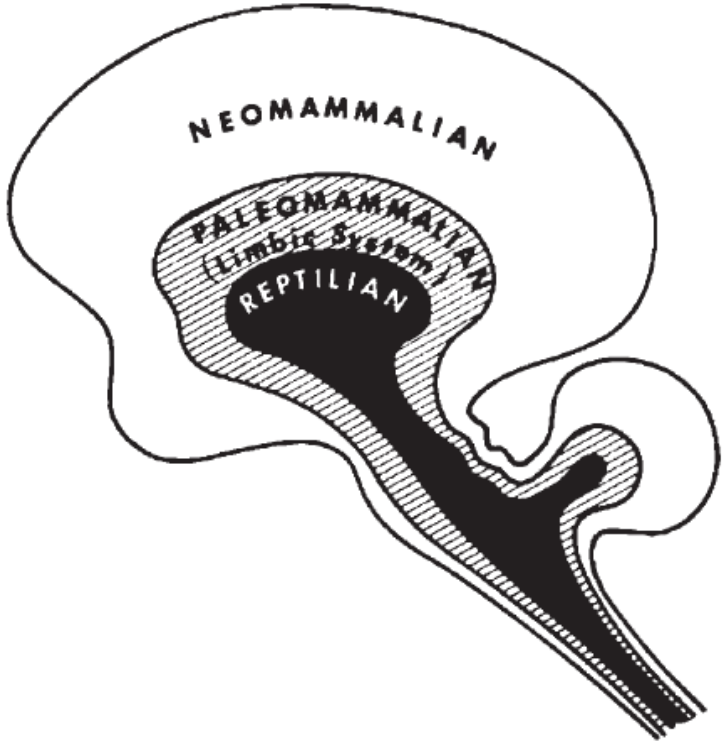


FIGURE 5.11 Diagram of the evolution of the mammalian brain. The forebrain evolves and expands along the lines of the three basic neural assemblies that anatomically and biochemically reflect ancestral commonalities with reptiles, early mammals, and late mammals. *Source*: Adapted from MacLean, 1967, with kind permission.

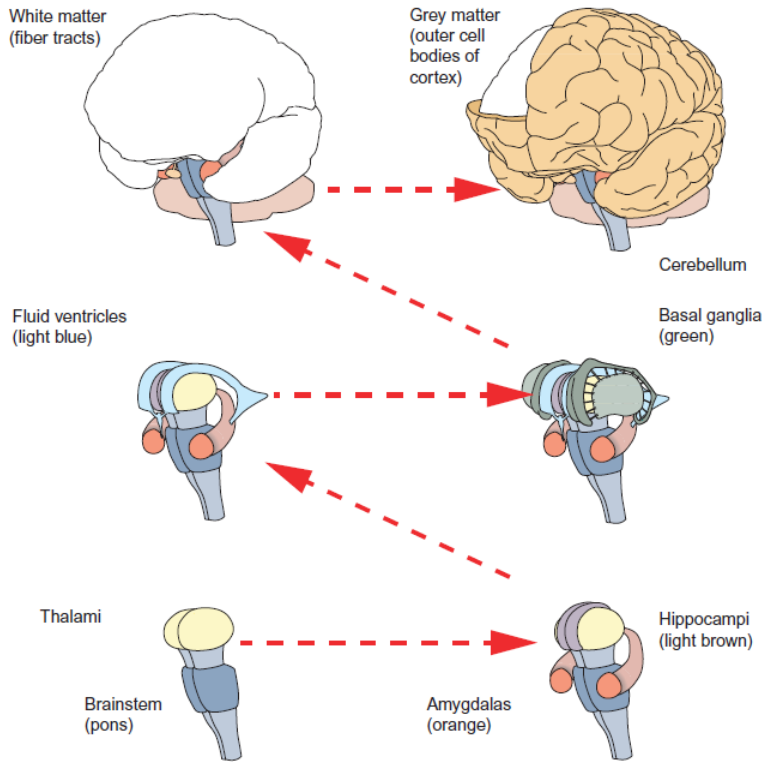


FIGURE 5.10 Growing the brain from the bottom up. If you can memorize these basic shapes, you will have a solid framework for understanding the brain. Notice how the brain builds on the brainstem, with the thalami on top as major input hub. The hippocampi and amygdalas are actually nestled inside each of the temporal lobes. The light blue fluid ventricles have no neurons, but provide the brain's own circulatory system. The basal ganglia can be thought of as the output hub of the system. A great deal of traffic flows back to the cortex as well. *Source*: Baars and Fu, with permission.

EL CEREBRO ES MAS QUE LA CORTEZA

TALLO CEREBRAL:

MEDULA, PUENTE, CEREBELO Y CEREBRO MEDIO

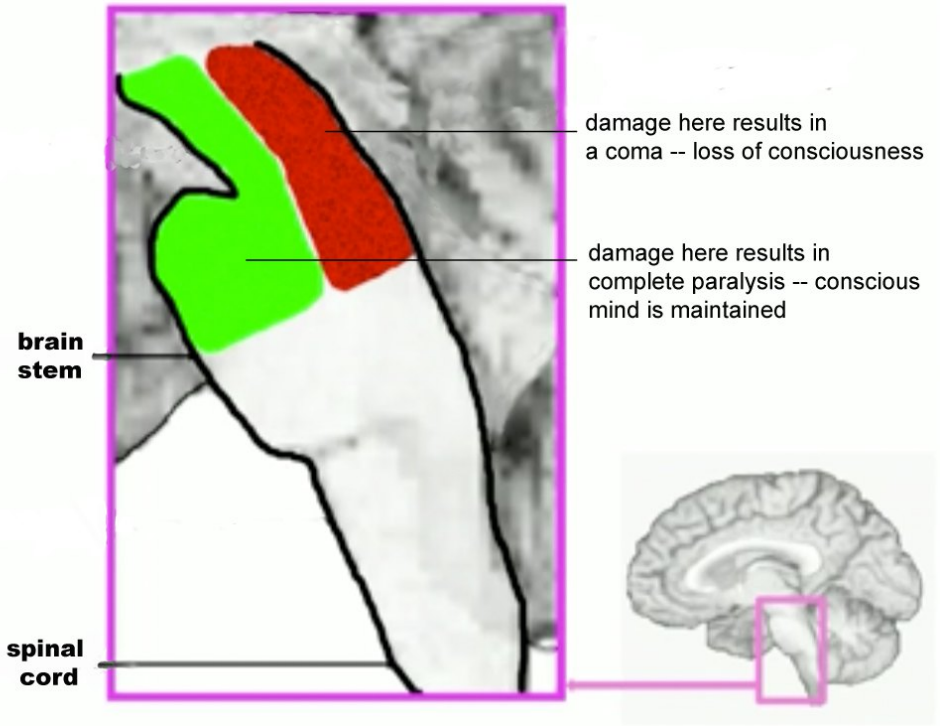


talamo

puente

medula

FIGURE 5.16 We begin 'growing' the brain with the brain-stem and pons. *Source*: Baars and Fu.



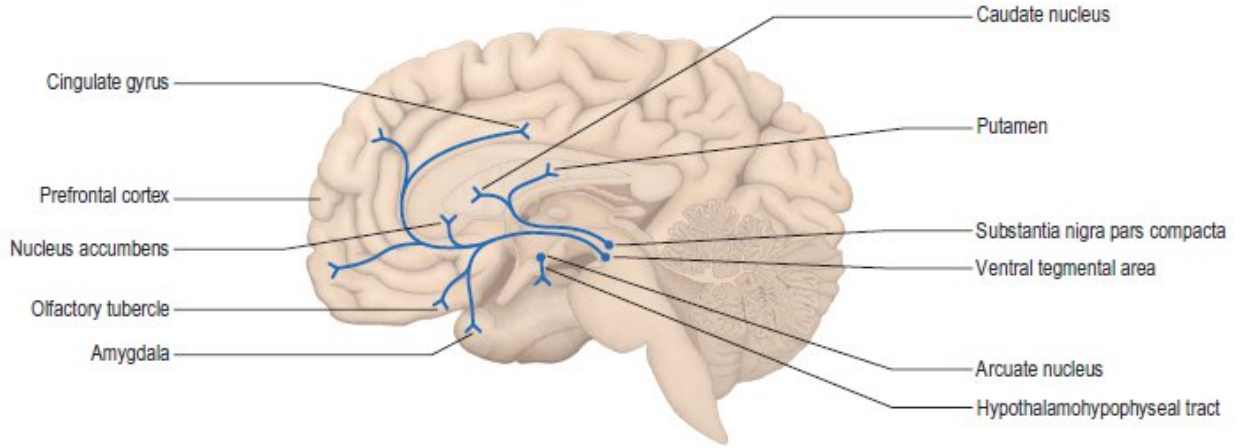


Figure 9.14 **Dopaminergic neurones and their principal projections.**

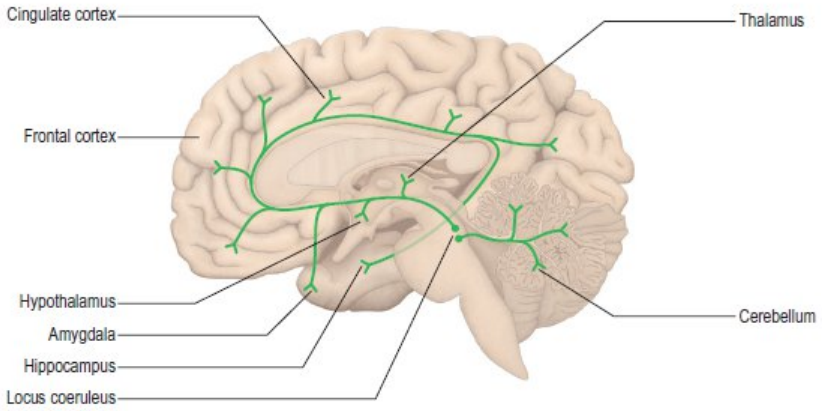
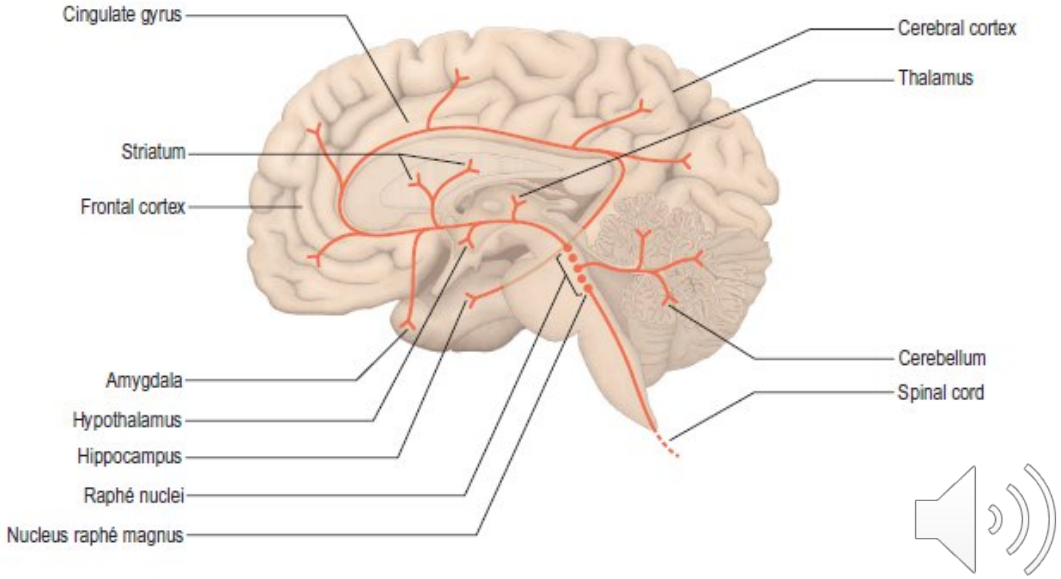


Figure 9.16 **Noradrenergic neurones and their principal projections.**



Serotonergic neurones and their principal projections.

Puente
cerebro cerebelo



FIGURE 5.16 We begin 'growing' the brain with the brainstem and pons. *Source*: Baars and Fu.

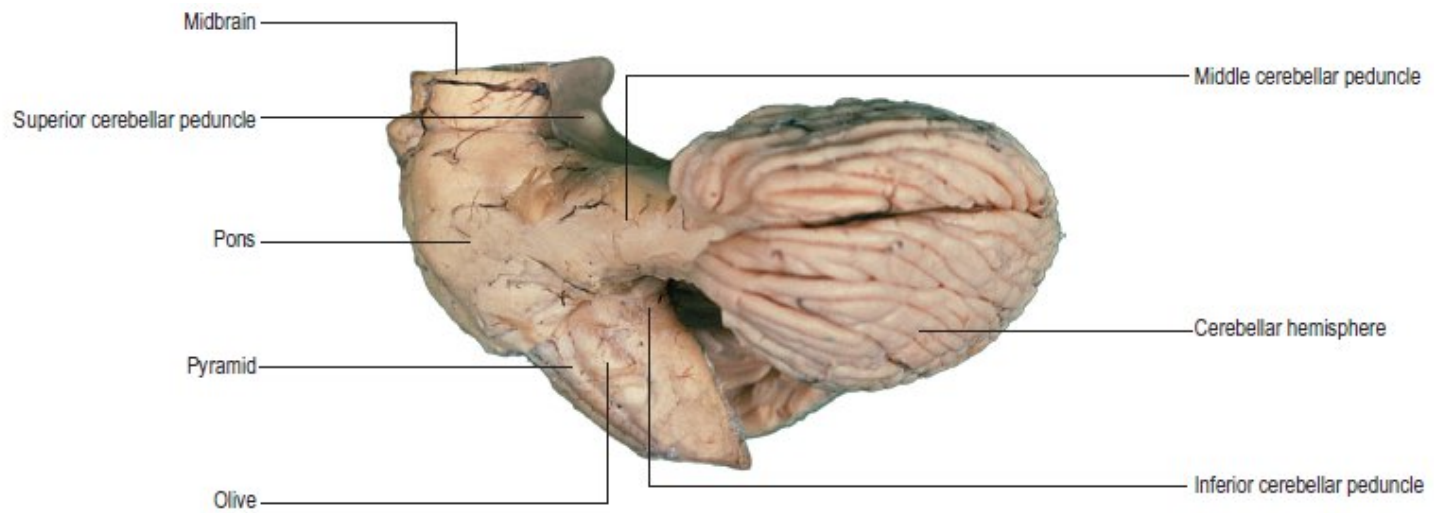


Figure 11.1 **Lateral aspect of the brainstem and cerebellum, showing the cerebellar peduncles.** Parts of the anterior, posterior and flocculonodular lobes have been removed to display the peduncles more clearly.

Nervios craneanos, formacion reticular

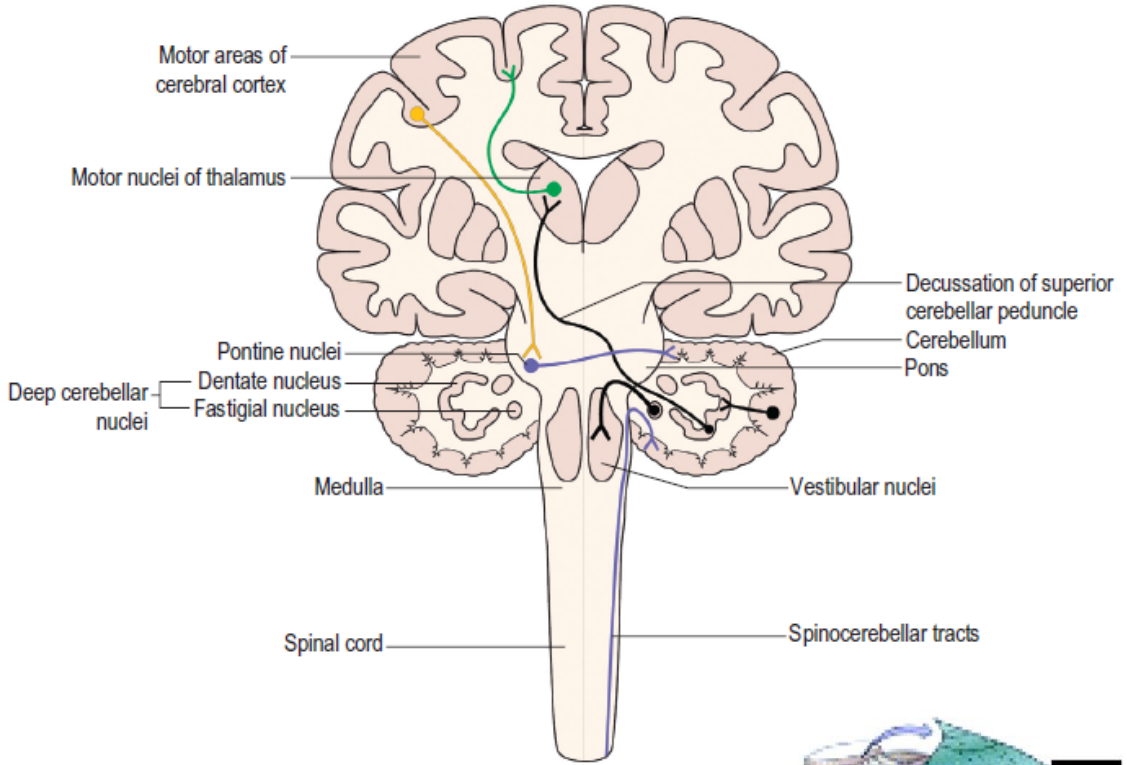
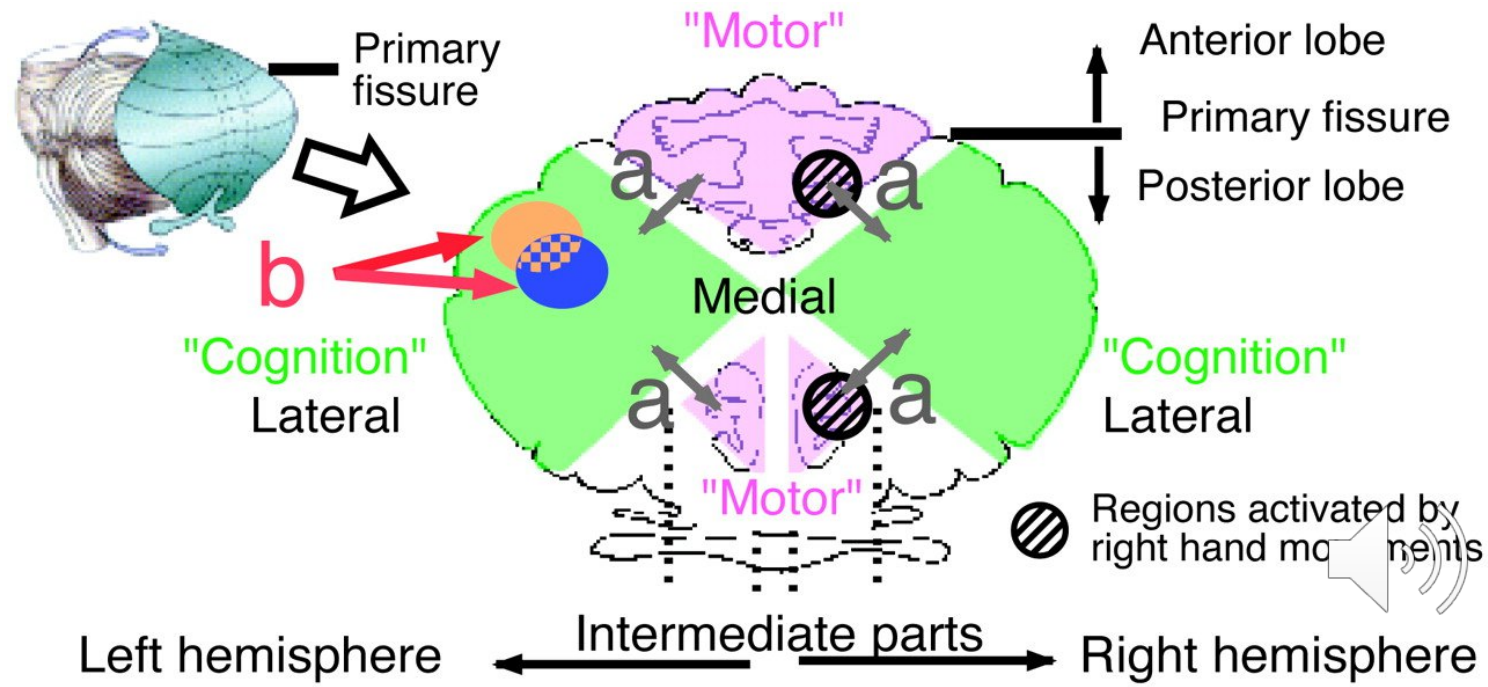


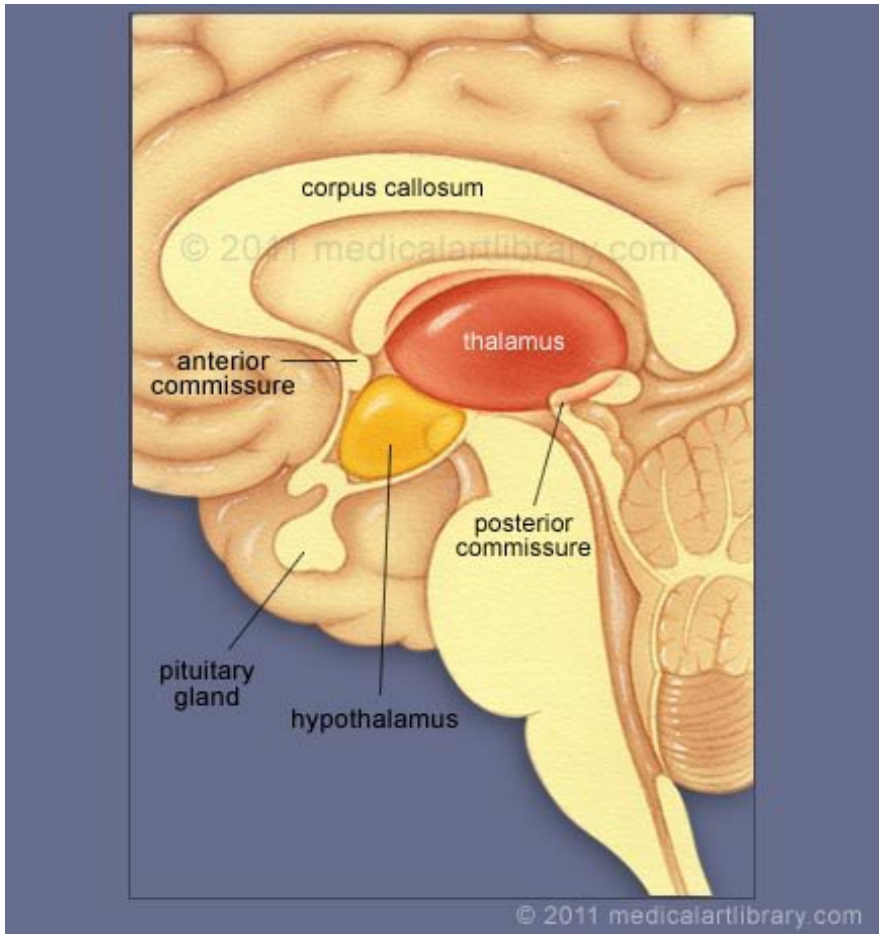
Figure 1.31 **Overview of the connections of the cerebellum.**



DIENCEFALO:
TALAMO- HIPOTALAMO



FIGURE 5.16 We begin 'growing' the brain with the brain-stem and pons. *Source*: Baars and Fu.



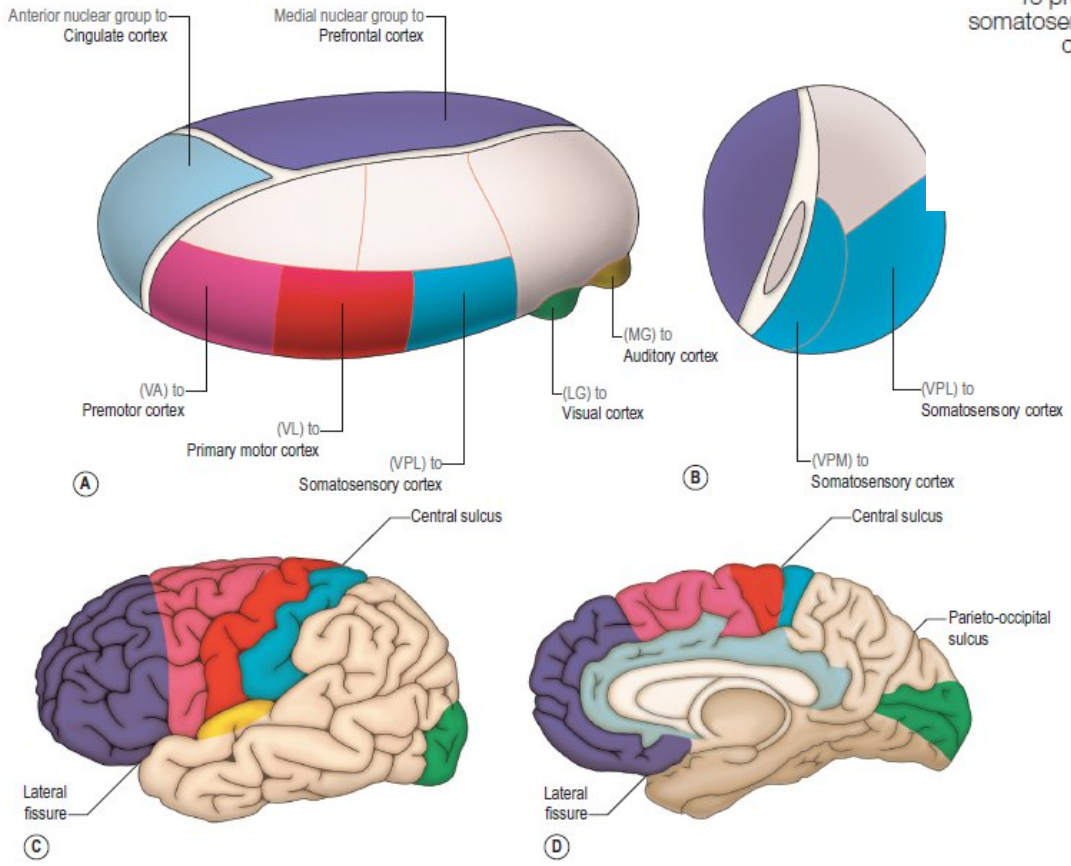


Figure 12.6 **Organisation of thalamic nuclei and their principal relationships with the cerebral cortex.** The thalamus is viewed from its anterolateral aspect **(A)** and in coronal section **(B)**. Colours indicate the relationships between thalamic nuclei and corresponding cerebral cortical regions on the lateral **(C)** and medial **(D)** aspects of the cerebral hemisphere.

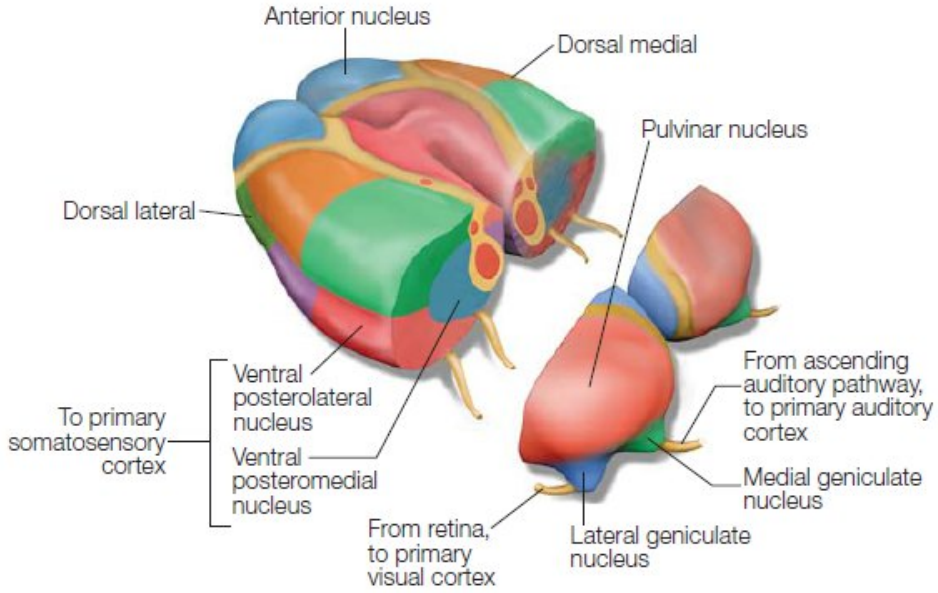
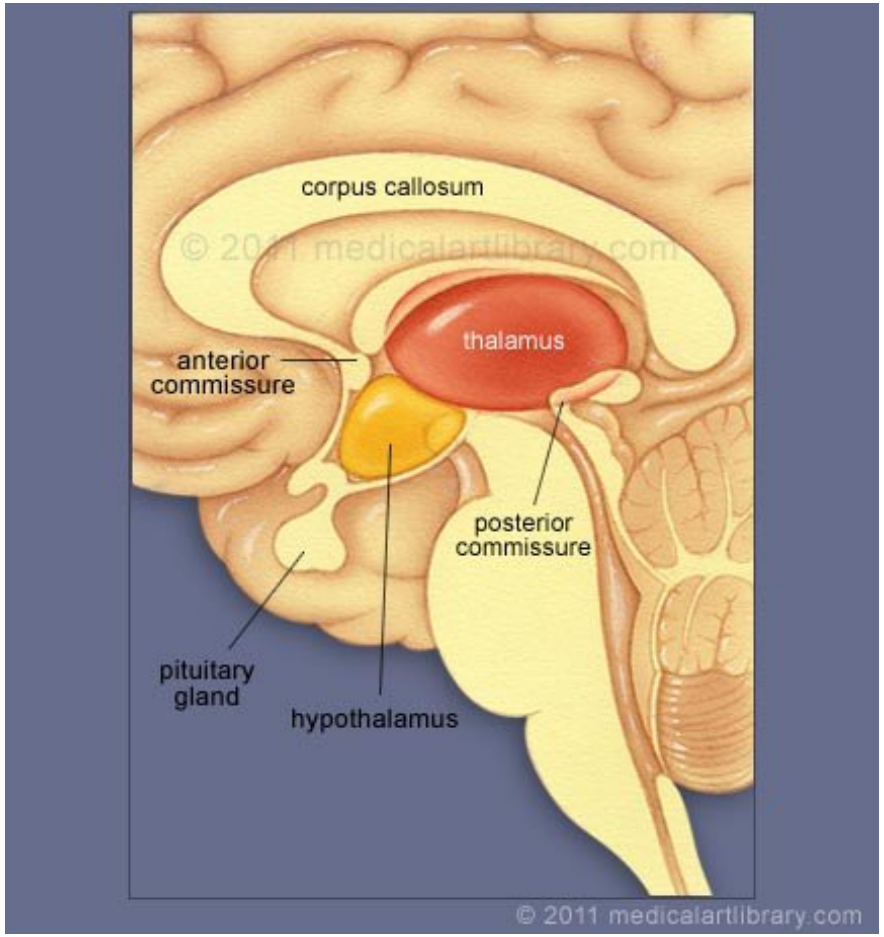


FIGURE 2.24 The thalamus, showing inputs and outputs and major subdivisions.
The various subdivisions of the thalamus serve different sensory systems and participate in various cortical–subcortical circuits. The posterior portion of the thalamus (lower right) is cut away in cross section and separated from the rest of the thalamus to reveal the internal organization of the thalamic nuclei (upper left).

HIPOTALAMO



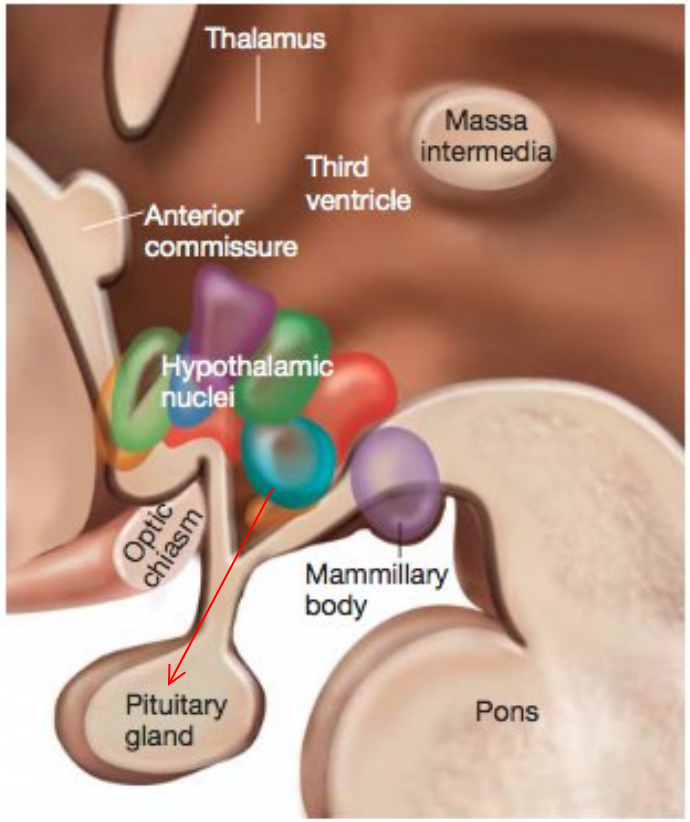


FIGURE 2.25 Midsagittal view of the hypothalamus.
Various nuclear groups are shown diagrammatically. The hypothalamus is the floor of the third ventricle, and, as the name suggests, it sits below the thalamus. Anterior is to the left in this drawing.

HOMEOSTASIS- ANS: SED, HAMBRE, FATIGA, TEMPERATURA, RITMOS CIRCADIANOS
SISTEMA ENDOCRINO (HORMONAS)

TELENCEFALO:

SISTEMA LIMBICO, GANGLIOS DE LA BASE Y CORTEX CEREBRAL

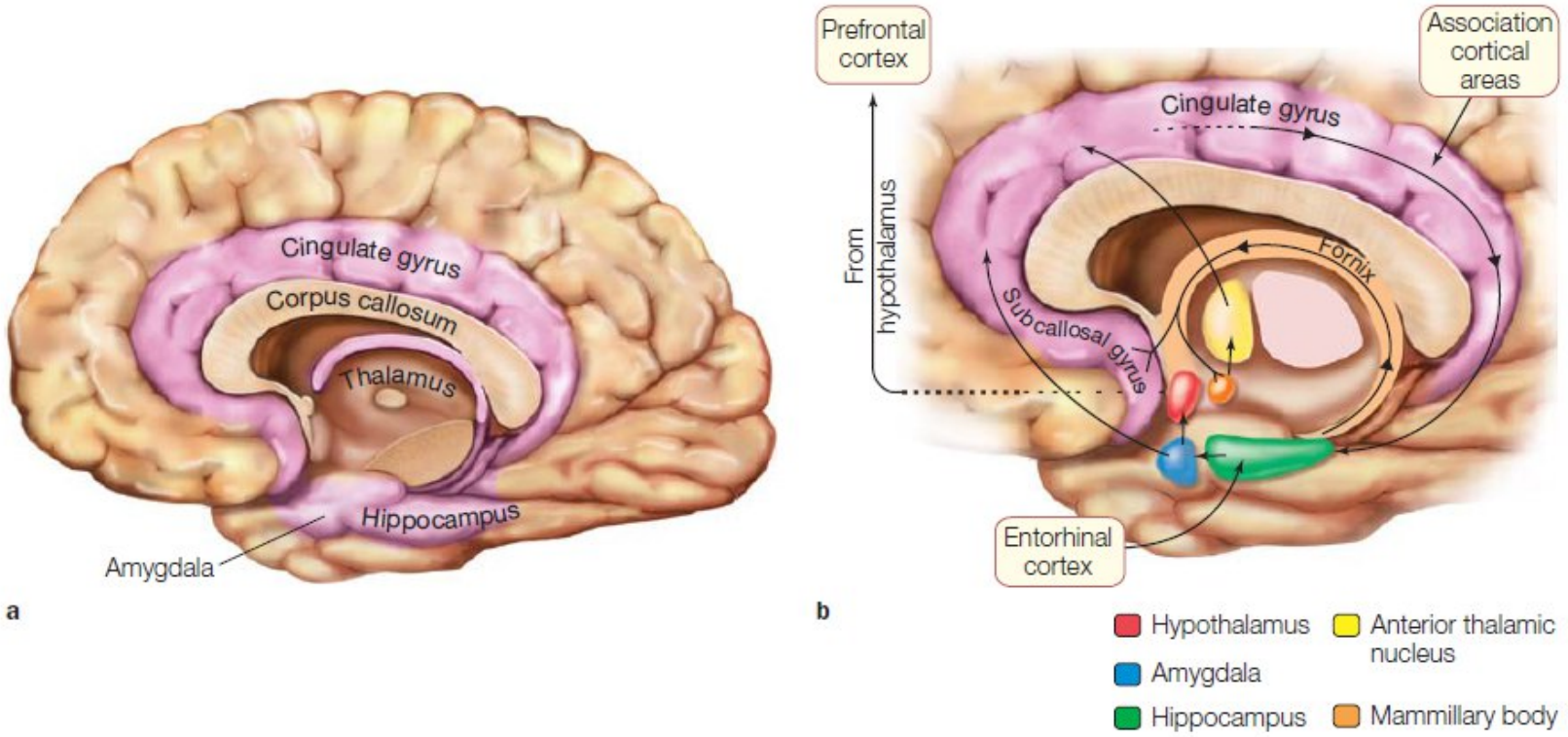


FIGURE 2.26 The human limbic system.
(a) Anatomy of the limbic system. (b) Major connections of the limbic system, shown diagrammatically in a medial view of the right hemisphere. The figure zooms in on the region in purple in (a). The basal ganglia are not represented in this figure, nor is the medial dorsal nucleus of the thalamus. More detail is shown here than needs to be committed to memory, but this figure provides a reference that will come in handy in later chapters.

SISTEMA LIMBICO (CEREBRIS LIMBUS)

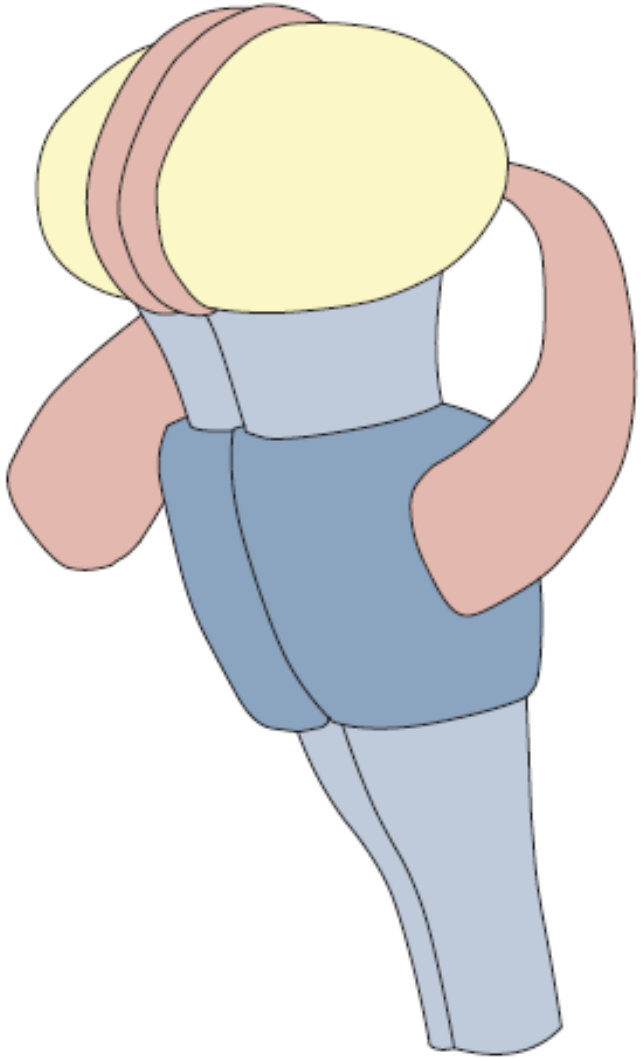


FIGURE 5.17 Schematic drawing of the hippocampi. *Source*: Baars and Fu.

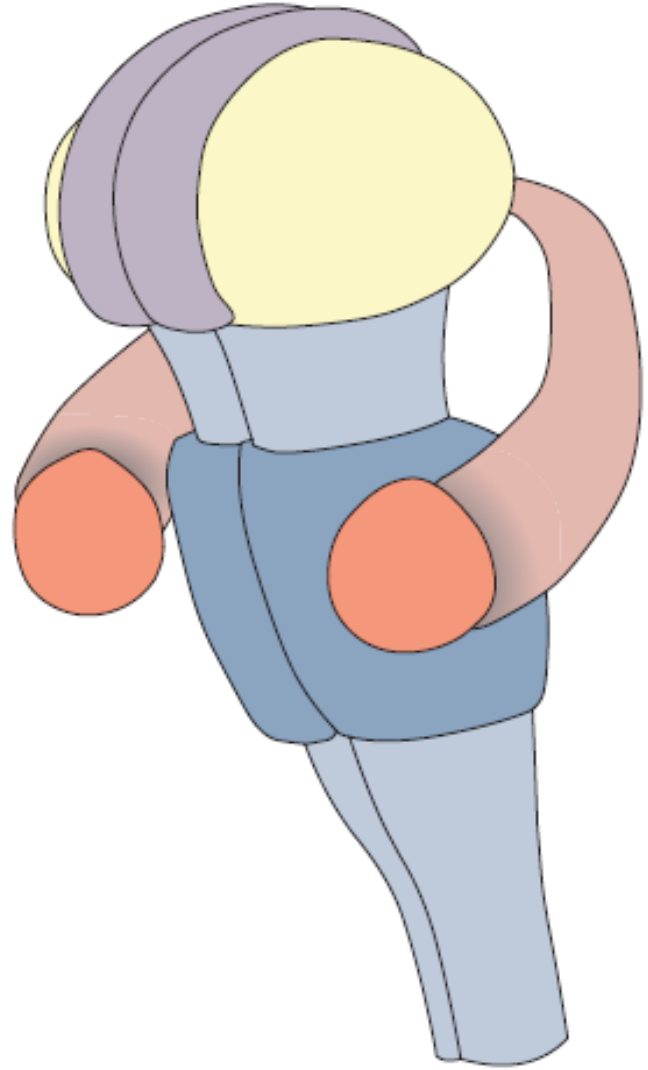
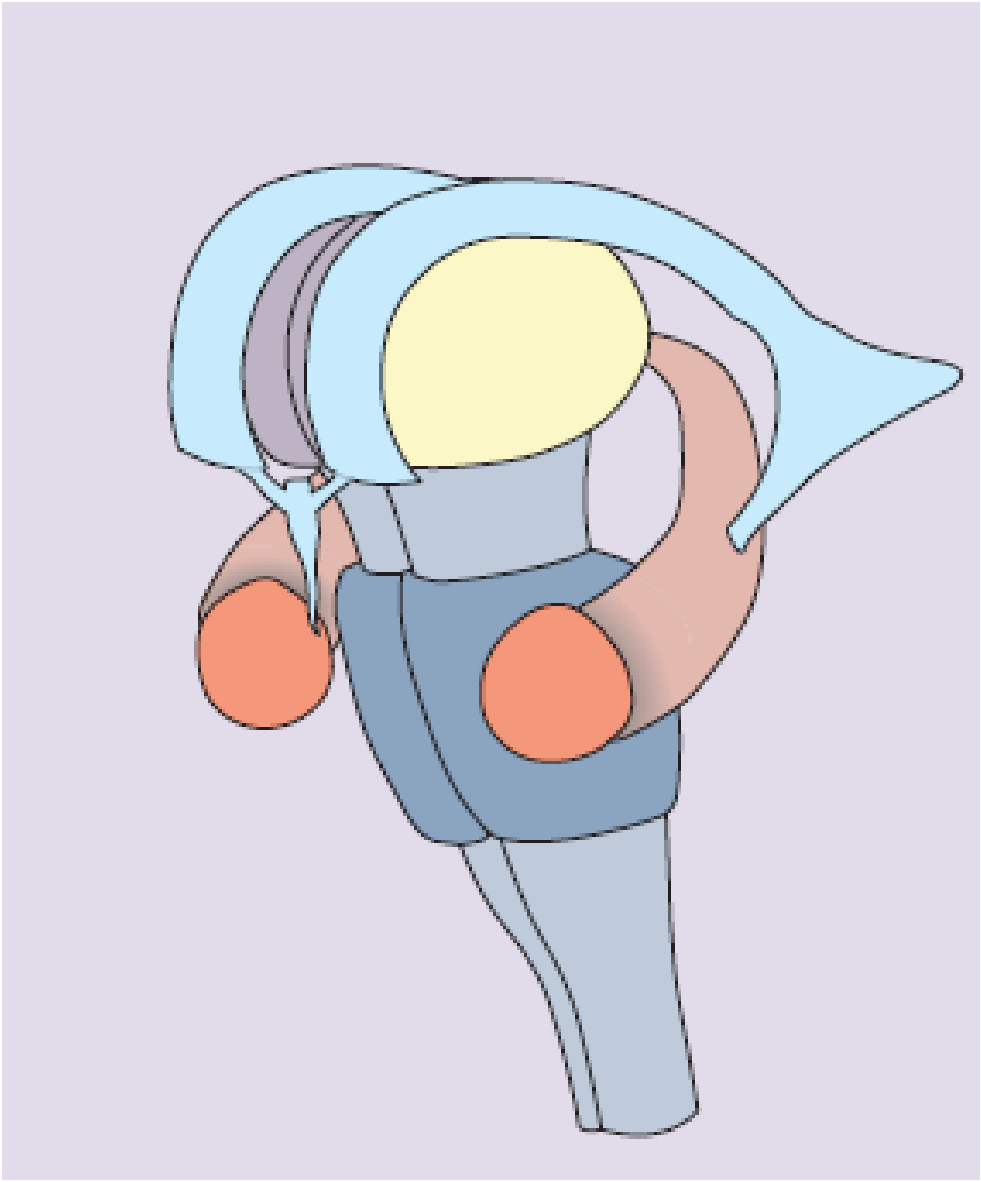


FIGURE 5.18 The amygdalas are situated just in front of the tip of each hippocampus. *Source*: Baars and Fu.



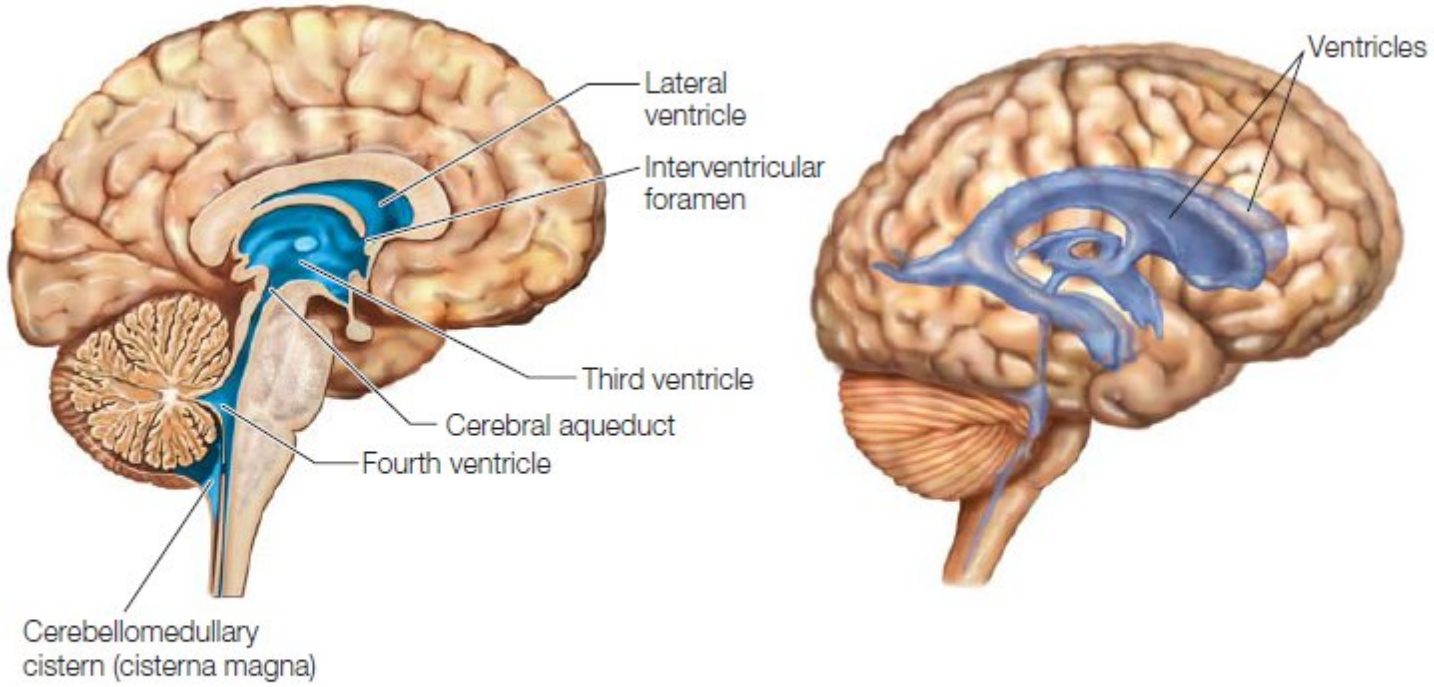
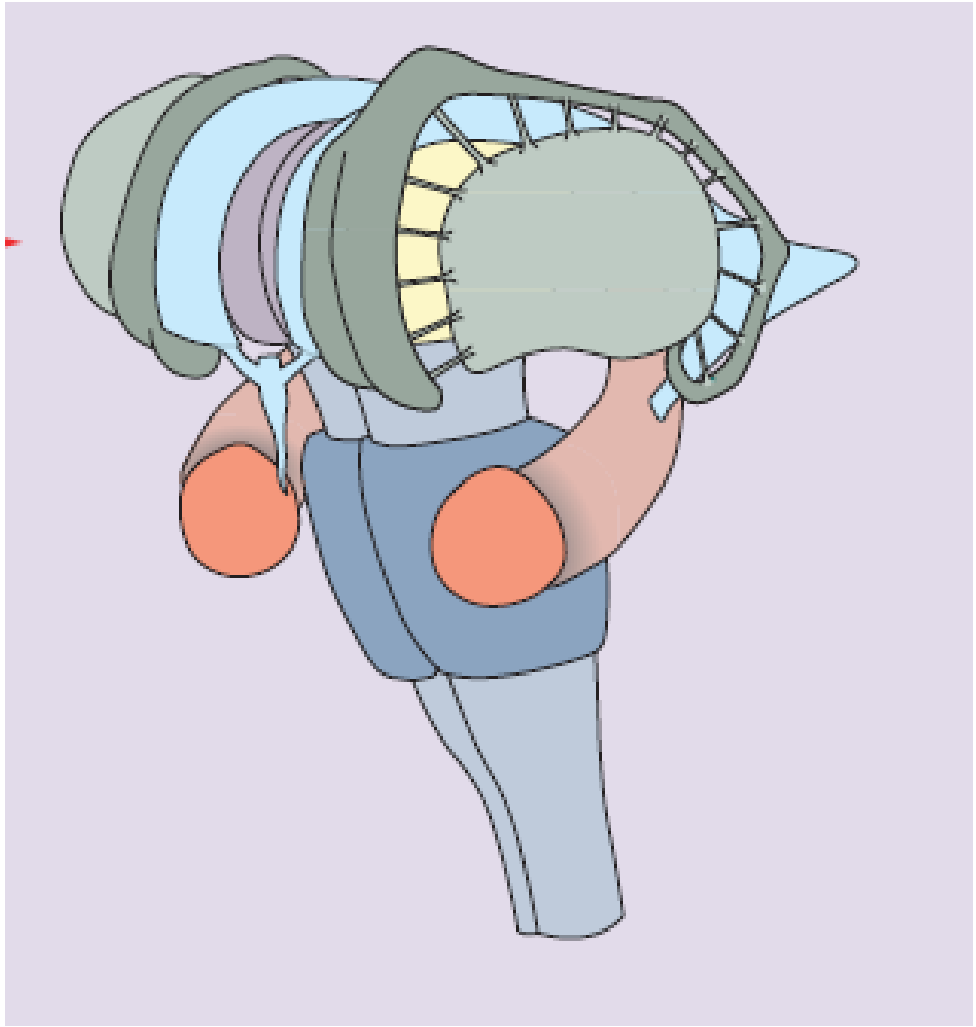


FIGURE 1 Ventricles of the human brain.
(left) Midsagital section showing the medial surface of the left hemisphere. (right) Transparent brain showing the ventricular system in 3D view.

GANGLIOS DE LA BA



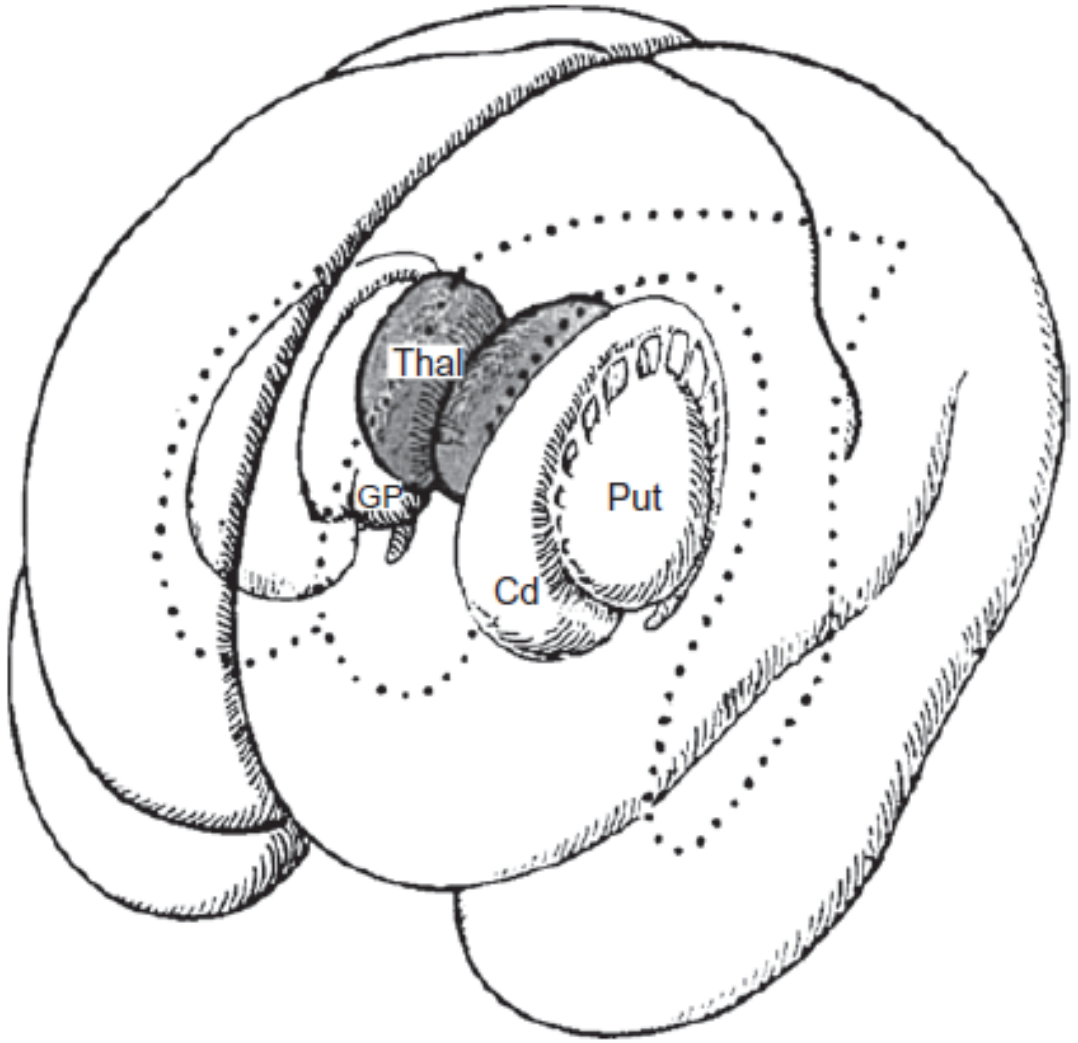
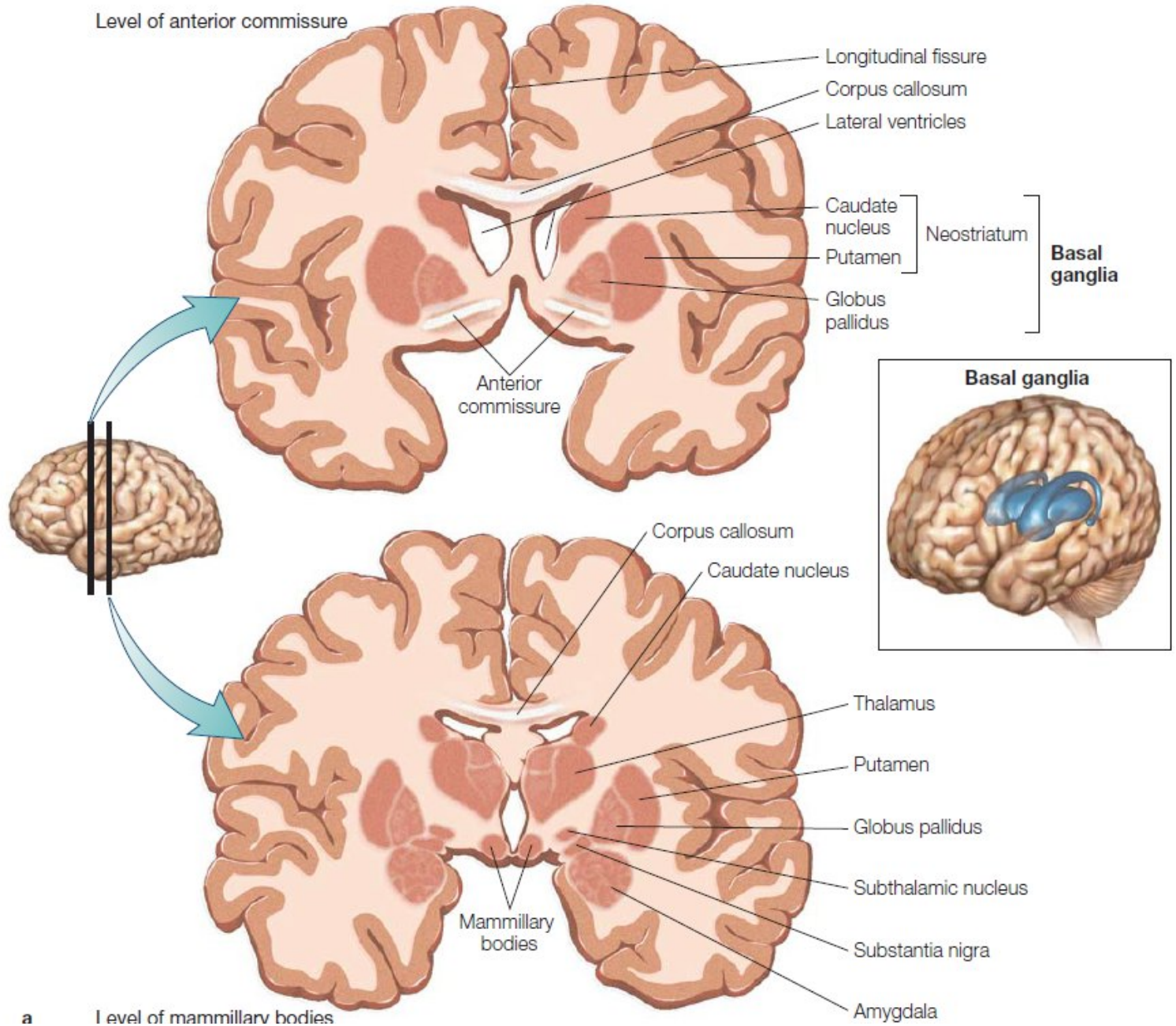
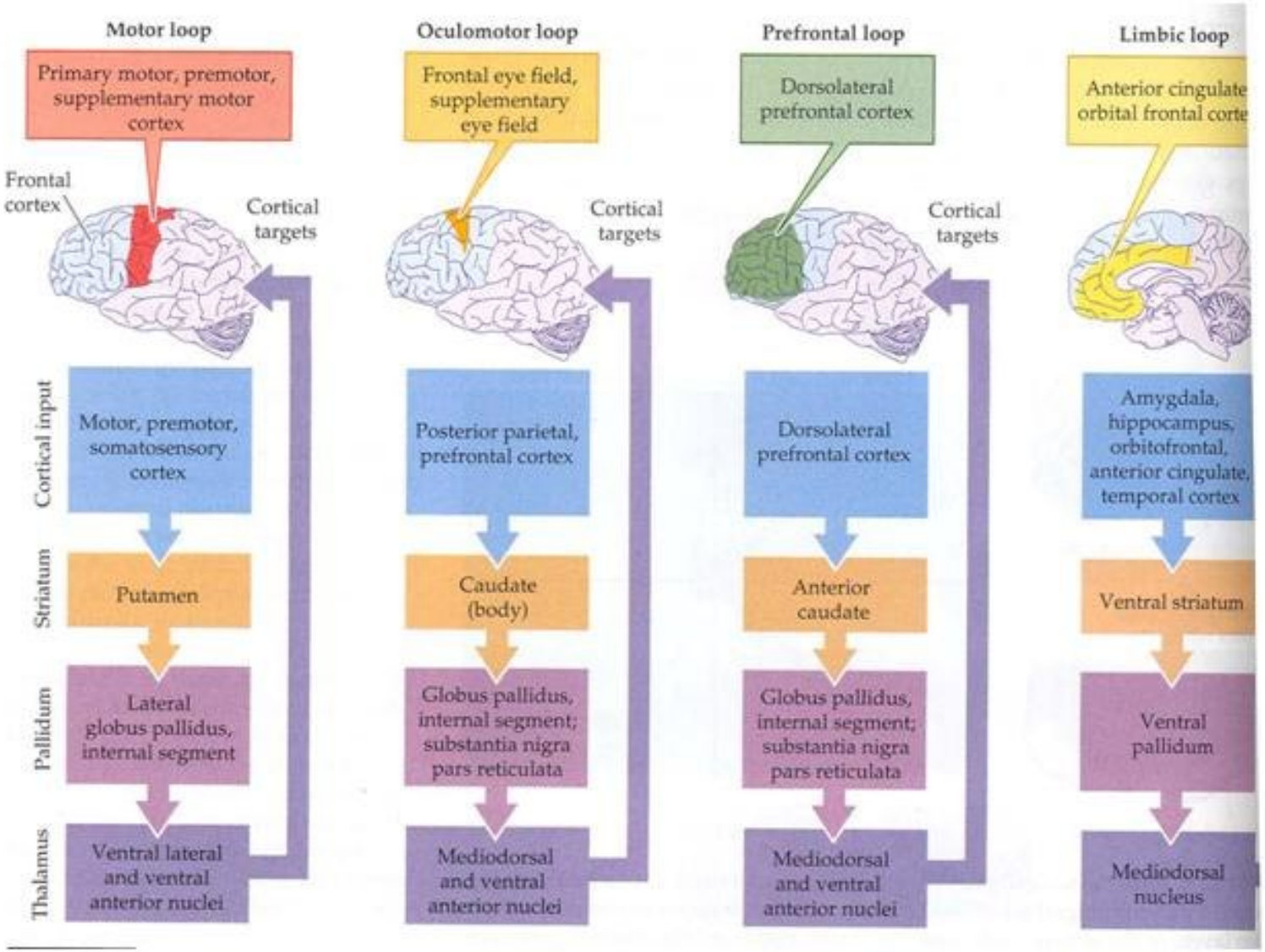


FIGURE 5.14 Transparent overview of the thalamus in the center of each hemisphere, and the basal ganglia looking like a 'shield and loop' on the outer side of each thalamus. *Source*: Ohye, 2002.


Level of anterior commissure
Longitudinal fissure
Corpus callosum
Lateral ventricles
Caudate nucleus
Putamen
Neostriatum
Basal ganglia
Globus pallidus
Anterior commissure
Basal ganglia
Corpus callosum
Caudate nucleus
Thalamus
Putamen
Globus pallidus
Subthalamic nucleus
Substantia nigra
Amygdala
Mammillary bodies
a
Level of mammillary bodies


Motor loop
Primary motor, premotor, supplementary motor cortex
Frontal cortex
Cortical targets
Cortical input
Motor, premotor, somatosensory cortex
Striatum
Putamen
Pallidum
Lateral globus pallidus, internal segment
Thalamus
Ventral lateral and ventral anterior nuclei
Oculomotor loop
Frontal eye field, supplementary eye field
Cortical targets
Posterior parietal, prefrontal cortex
Caudate (body)
Globus pallidus, internal segment; substantia nigra pars reticulata
Mediodorsal and ventral anterior nuclei
Prefrontal loop
Dorsolateral prefrontal cortex
Cortical targets
Dorsolateral prefrontal cortex
Anterior caudate
Globus pallidus, internal segment; substantia nigra pars reticulata
Mediodorsal and ventral anterior nuclei
Limbic loop
Anterior cingulate orbital frontal corte
Amygdala, hippocampus, orbitofrontal, anterior cingulate, temporal cortex
Ventral striatum
Ventral pallidum
Mediodorsal nucleus

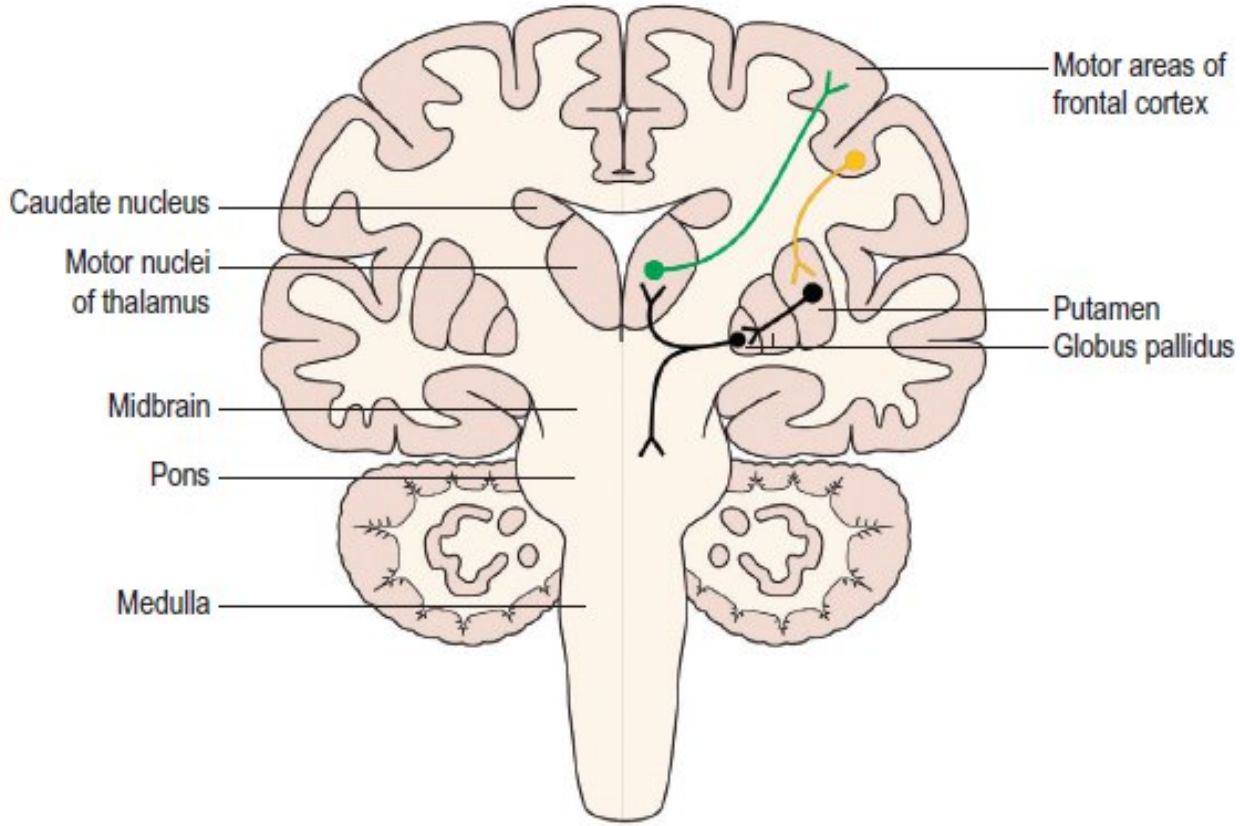


Figure 1.30 **Overview of the connections of the basal ganglia.**

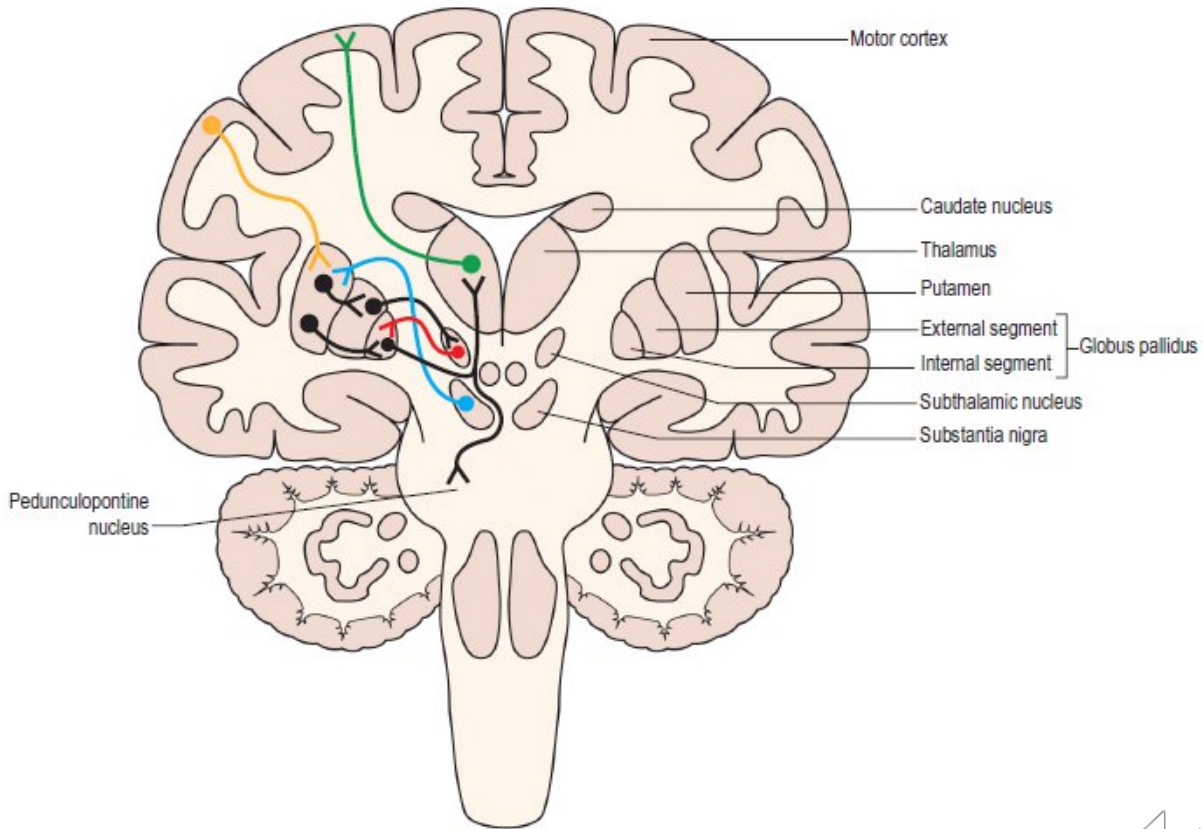
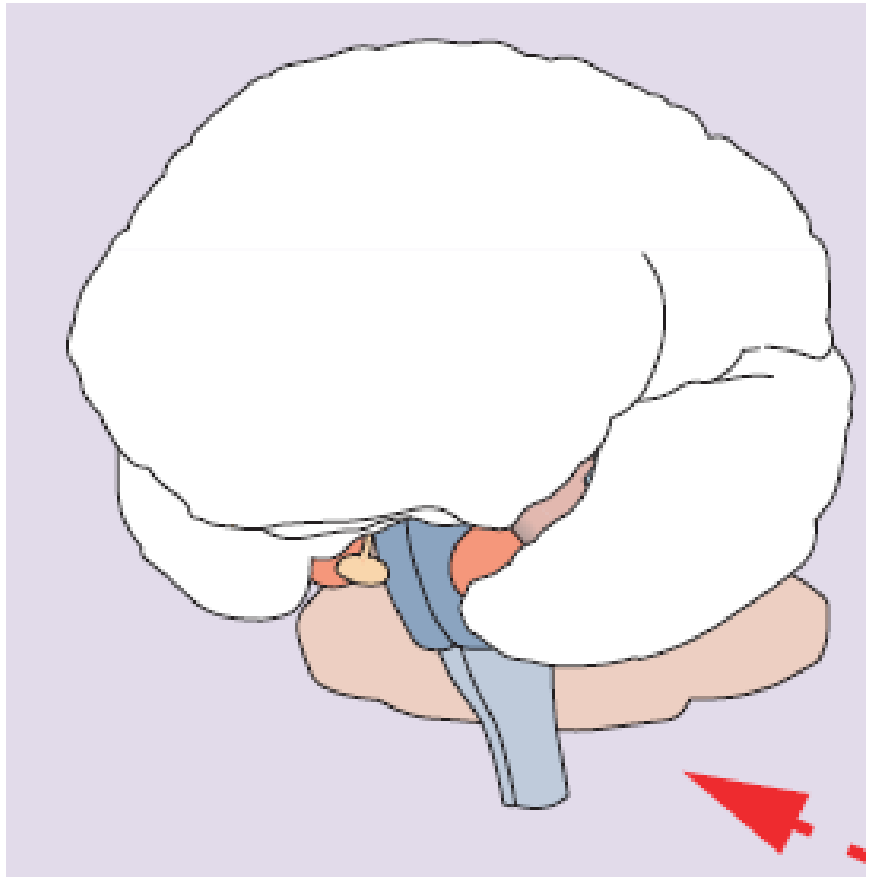


Figure 14.5 **Schematic diagram illustrating the principal connections of the corpus striatum and related nuclei.** Afferents to the striatum from the intralaminar thalamic nuclei and raphé nuclei have been omitted. For the sake of clarity, all efferents from the basal ganglia system are shown to originate from the medial segment of the globus pallidus, those from the pars reticulata of the substantia nigra being omitted. Similarly, efferents from the striatum are shown to originate only from the putamen and not from the caudate nucleus.

CORTEZA CEREBRAL



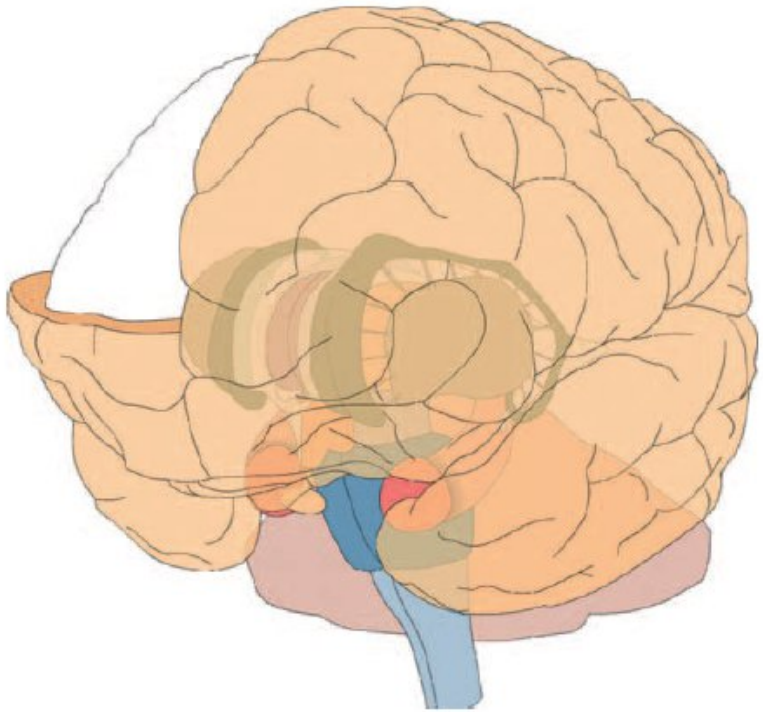
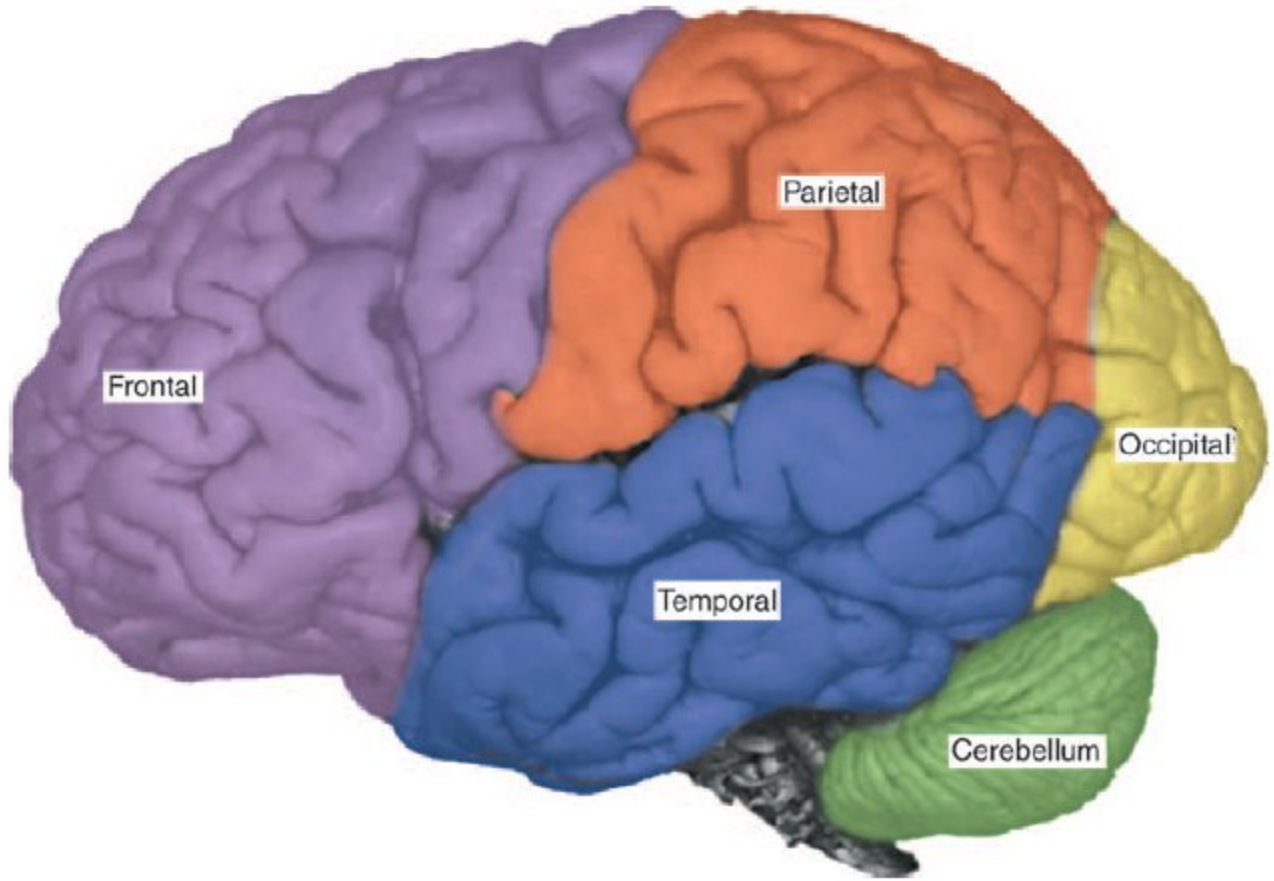
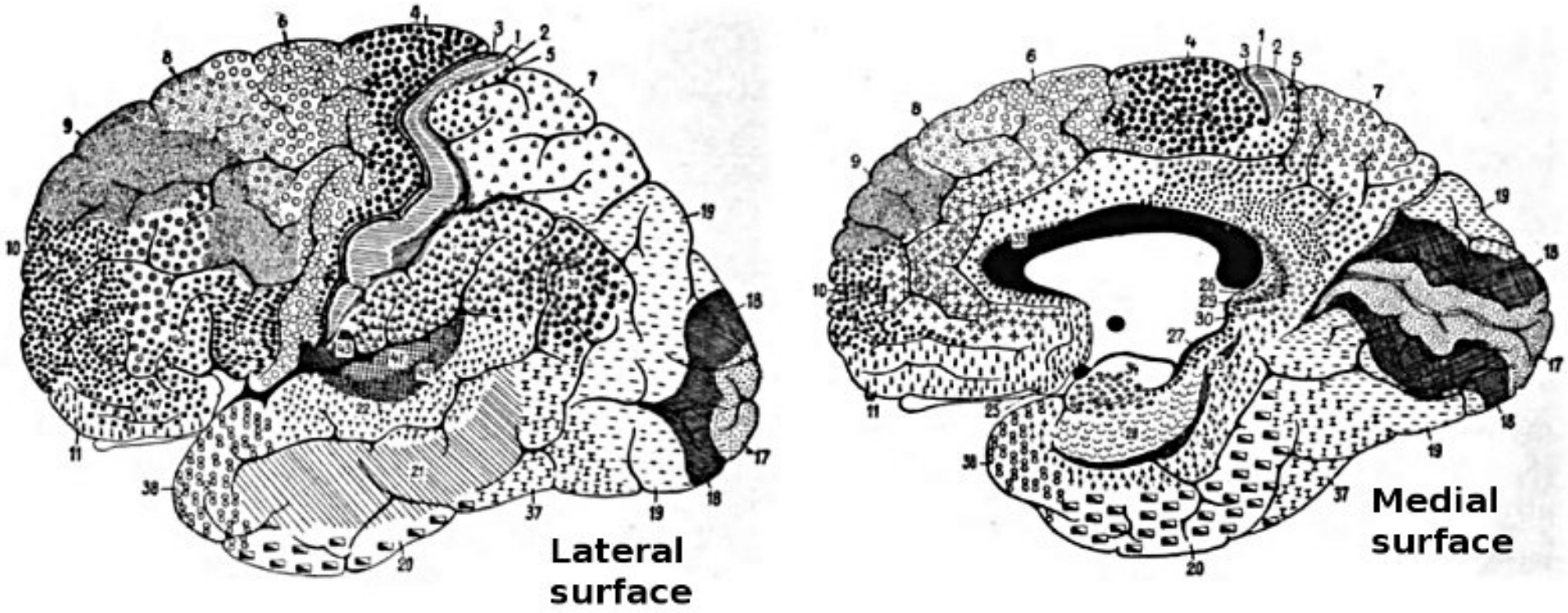


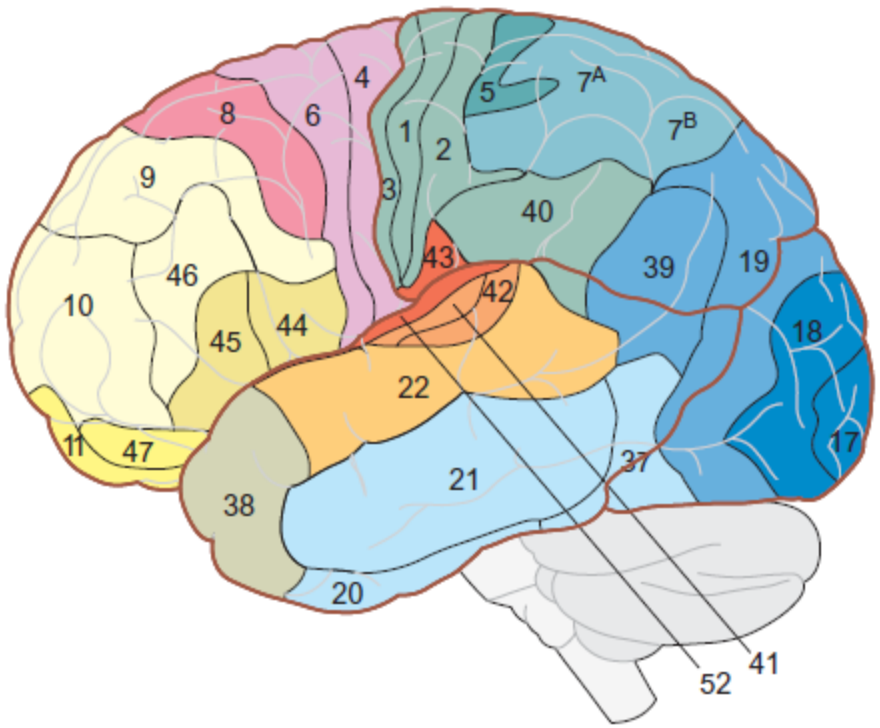
FIGURE 5.20 The cerebral hemispheres are shown mounted above the brainstem and other subcortical bodies. *Source*: Baars and Fu.


Parietal
Frontal
Occipital
Temporal
Cerebellum

CORTEZA CEREBRAL
ORGANIZACION CITOARQUITECTONICA



DIFERENCIAS EN MORFOLOGIA Y ORGANIZACION CELULAR



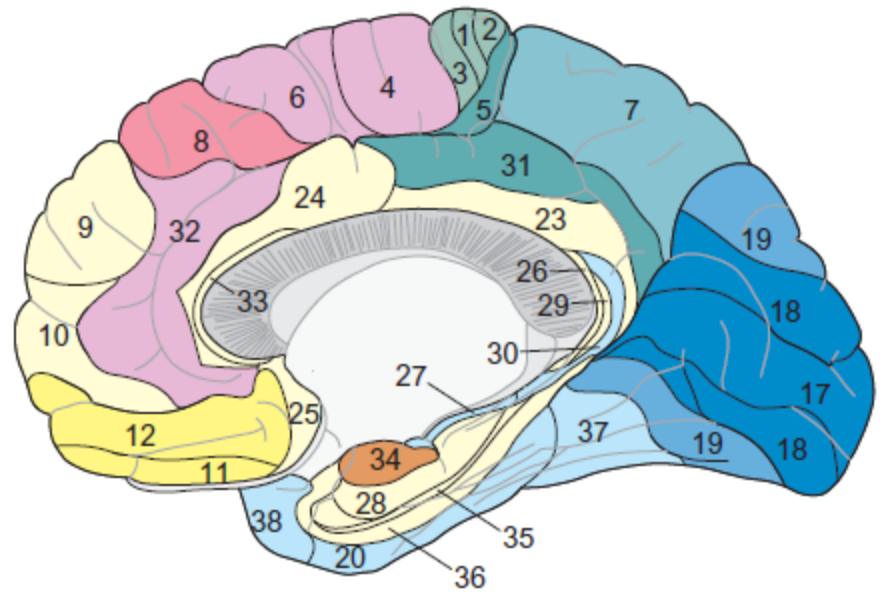


FIGURE 5.7 The Brodmann classification of regions in the right hemisphere, shown in a midsagittal view.

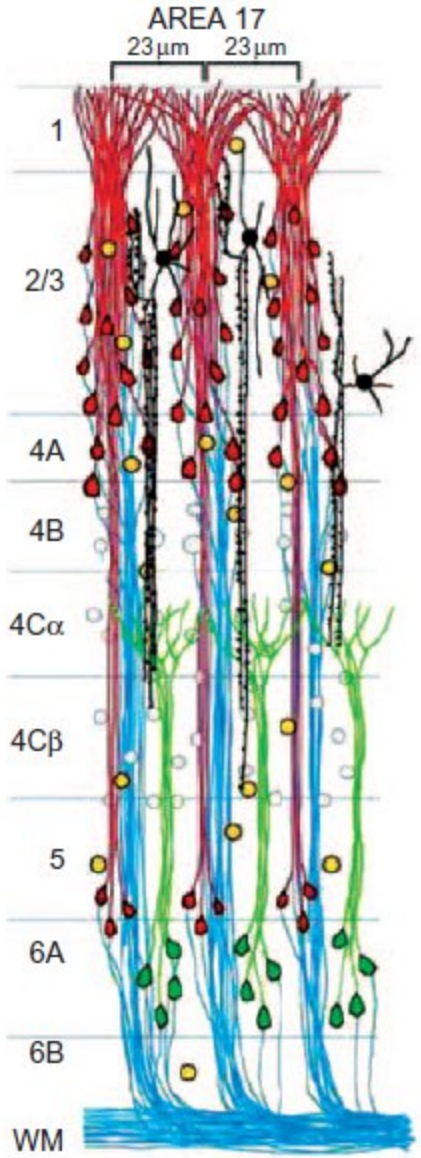


FIGURE 5.8 The six major layers of cortex in cross section. The figure shows three columns in Area 17, also called V1, the first visual projection area to the cortex. *Source*: Squire *et al.*, 2003.

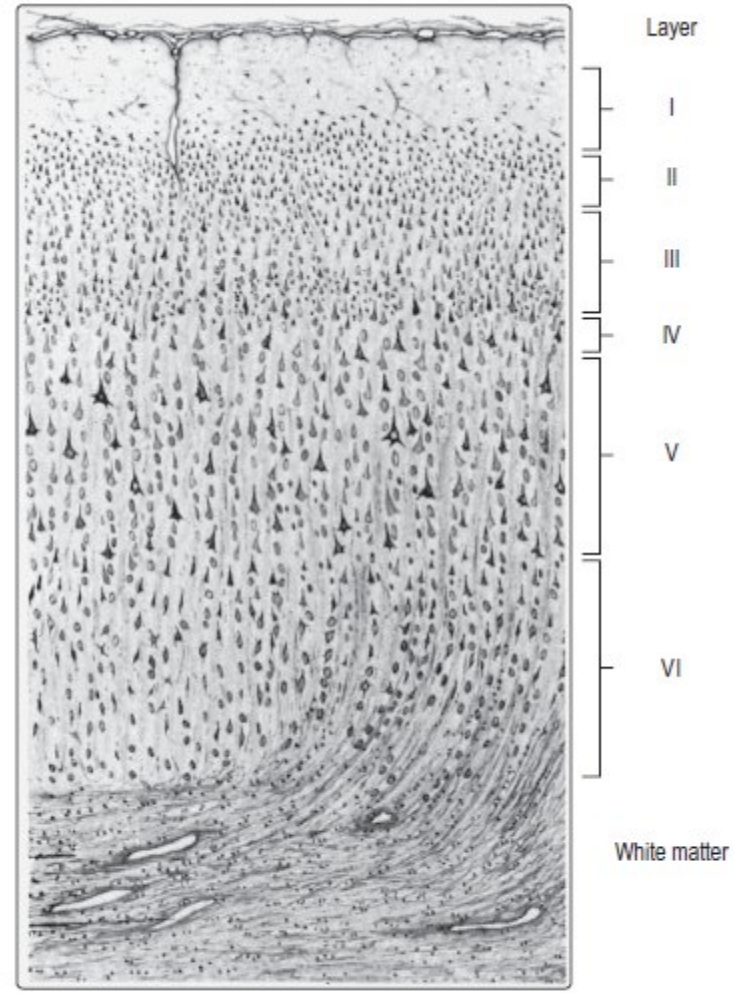


Figure 13.19 **The histological structure of the cerebral cortex.** (From Mitchell, G A G and Patterson, E L. Basic anatomy. London: Livingstone; 1954. Courtesy of Churchill Livingstone.)

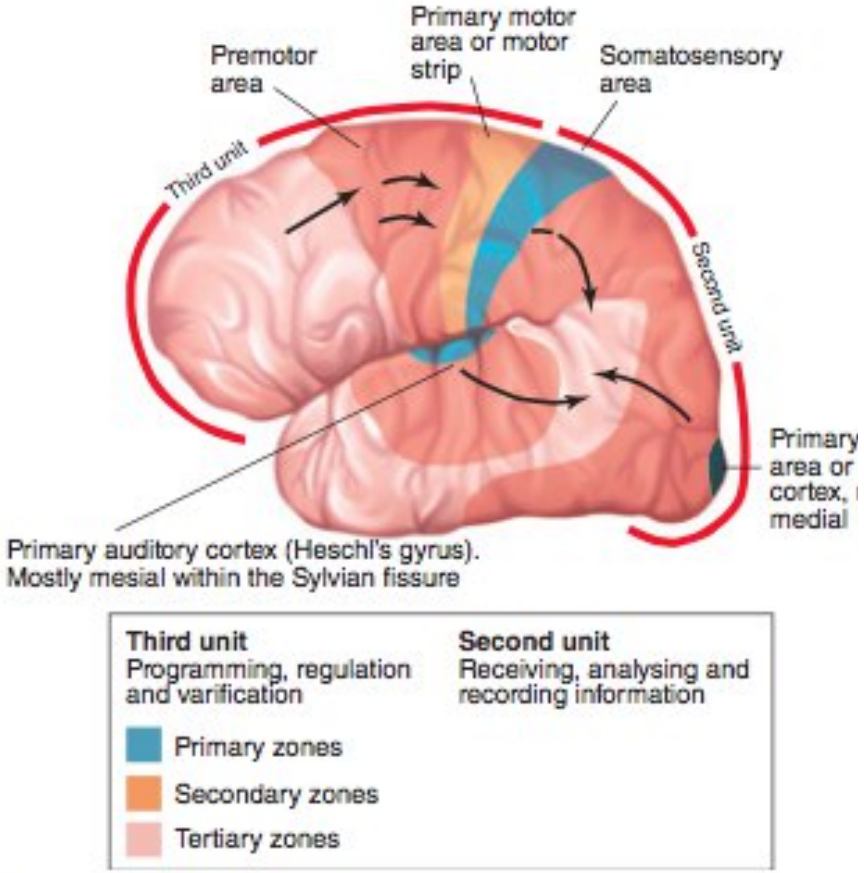


Figure 1.13 A diagram of the second unit for receiving analysing and storing information and the unit for the mo involvement of the third unit for the programming, regulat and verification of activity (adapted from Luria, 1973). No that the flow for the unit receiving information is in the re direction compared to the programming of activity.

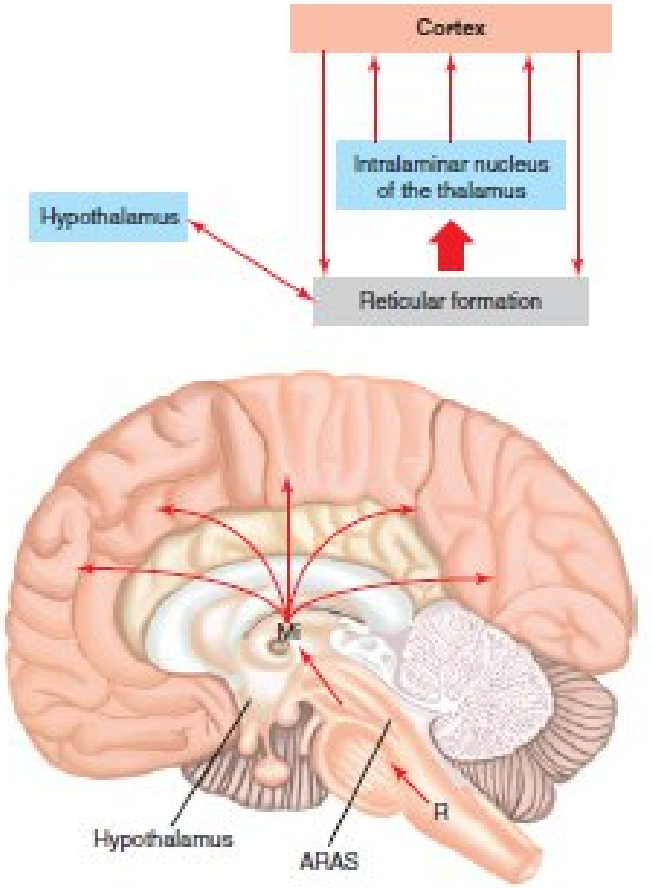
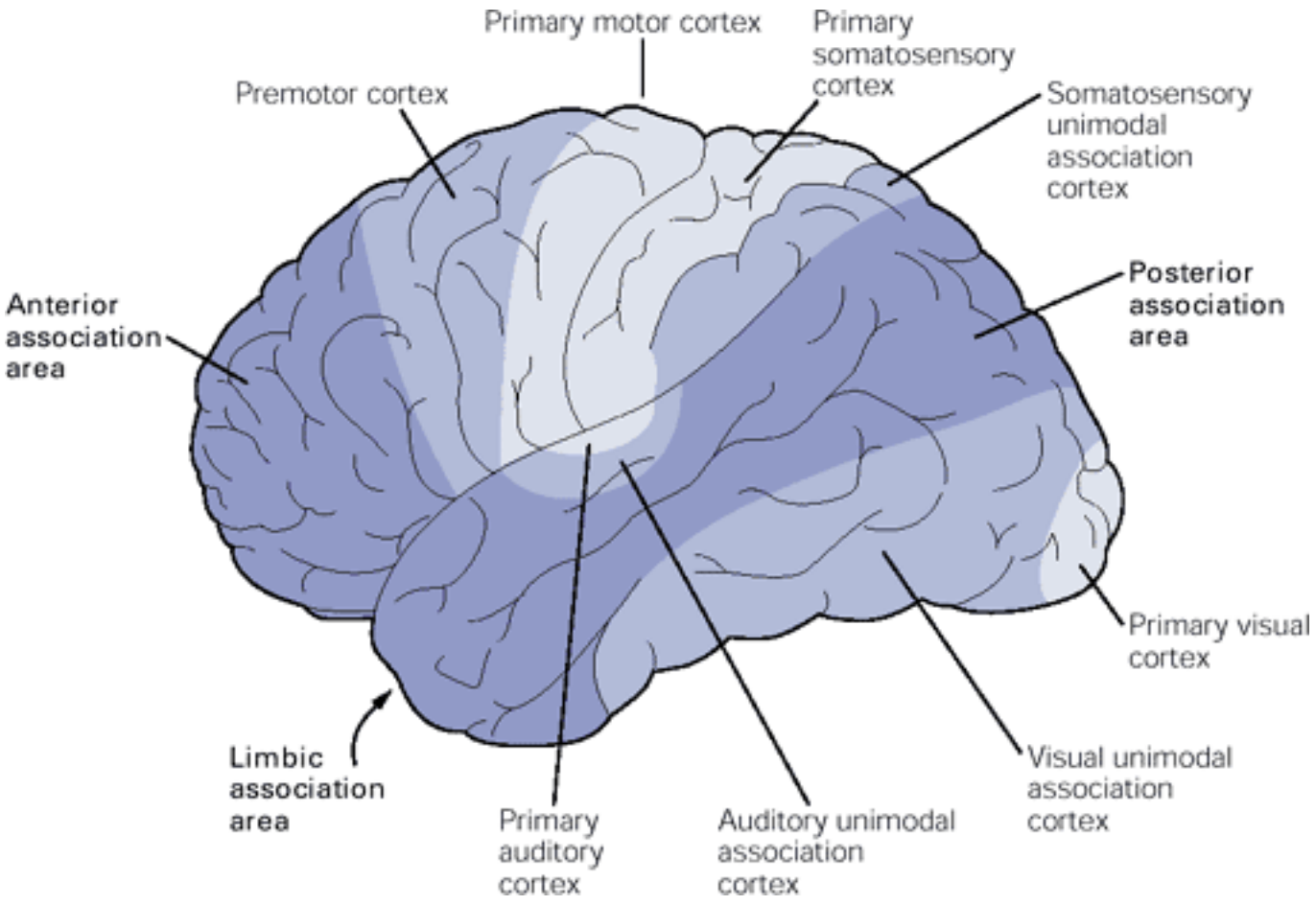


Figure 1.12 A graphic illustration of the reticular formation and the ascending reticular activating system (ARAS) pathways including the important pathway from the reticular formation to the intralaminar nuclei of the thalamus and then onto the cortices. Pathways exist also to an area within the hypothalamus. There are also descending influences from the hypothalamus on the reticular formation and from the cortex, especially from sensory areas. These descending pathways influence the reticular formation to maintain and dampen level of excitation in the ARAS, thereby allowing an optimal level of cortical arousal.

ASSOCIATION CORTEX



UNIMODAL Y HETEROMODAL

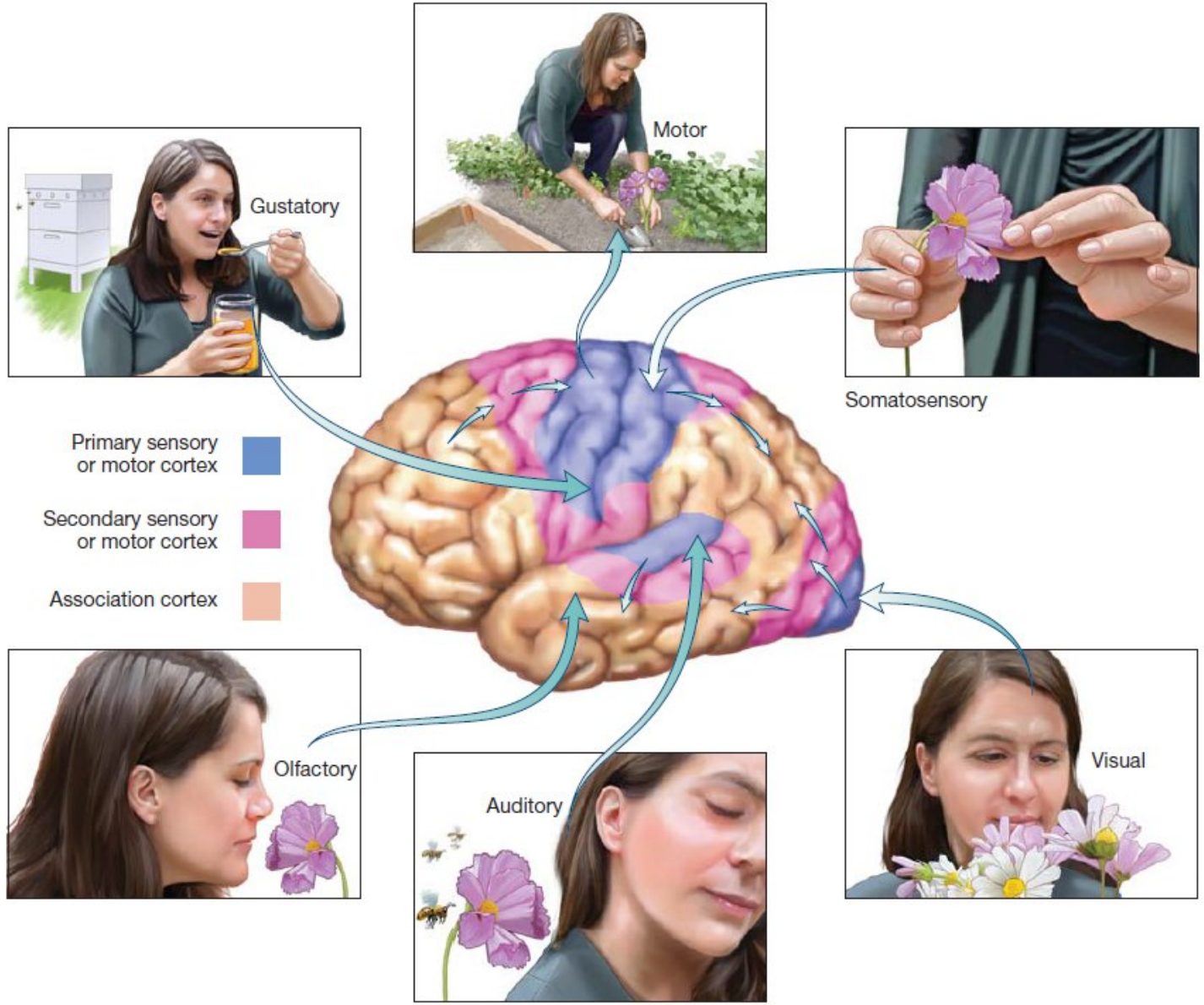


FIGURE 2.37 Primary sensory and motor cortex and surrounding association cortex.
The blue regions show the primary cortical receiving areas of the ascending sensory pathways and the primary output region to the spinal cord. The secondary sensory and motor areas are colored pink. The remainder is considered association cortex.

Anatomy and Functional Areas of the Brain

Functional Areas of the Cerebral Cortex

1 Visual Area:
- Sight
- Image recognition
- Image perception

2 Association Area
- Short-term memory
- Equilibrium
- Emotion

3 Motor Function Area
- Initiation of voluntary muscles

4 Broca's Area
- Muscles of speech

5 Auditory Area
- Hearing

6 Emotional Area
- Pain
- Hunger
- "Fight or flight" response

7 Sensory Association Area

8 Olfactory Area
- Smelling

9 Sensory Area
- Sensation from muscles and skin

10 Somatosensory Association Area
- Evaluation of weight, texture, temperature, etc. for object recognition

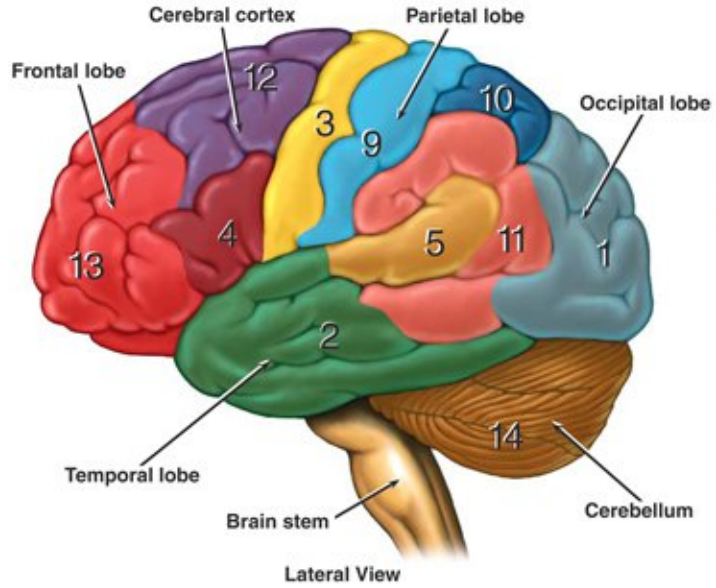
11 Wernicke's Area
- Written and spoken language comprehension

12 Motor Function Area
- Eye movement and orientation

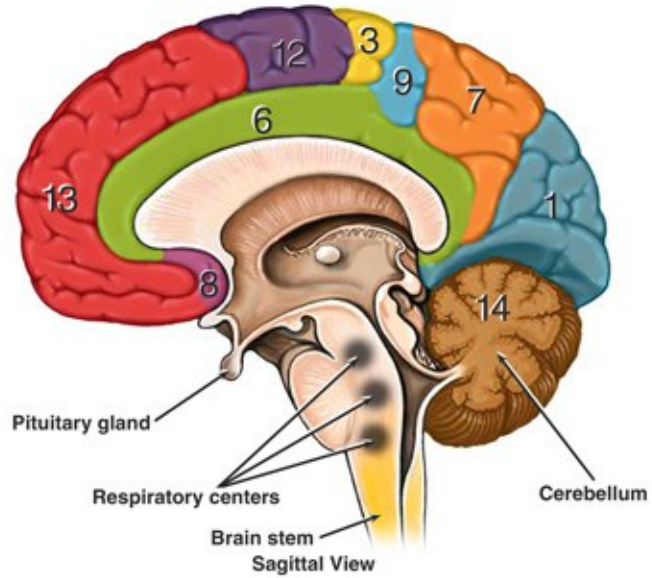
13 Higher Mental Functions
- Concentration
- Planning
- Judgment
- Emotional expression
- Creativity
- Inhibition

Functional Areas of the Cerebellum

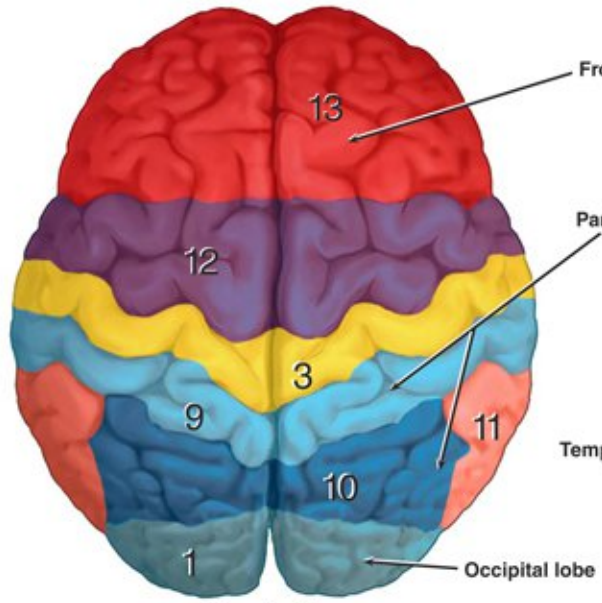
14 Motor Functions
- Coordination of movement
- Balance and equilibrium
- Posture



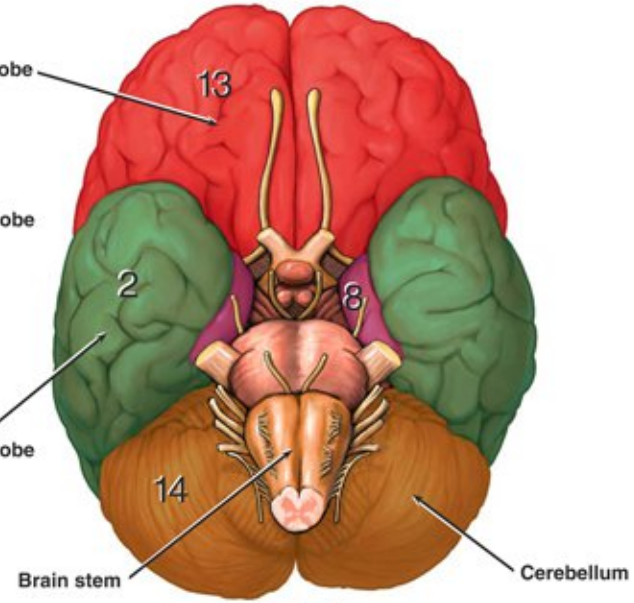
Lateral View



Sagittal View



Superior View



Inferior View

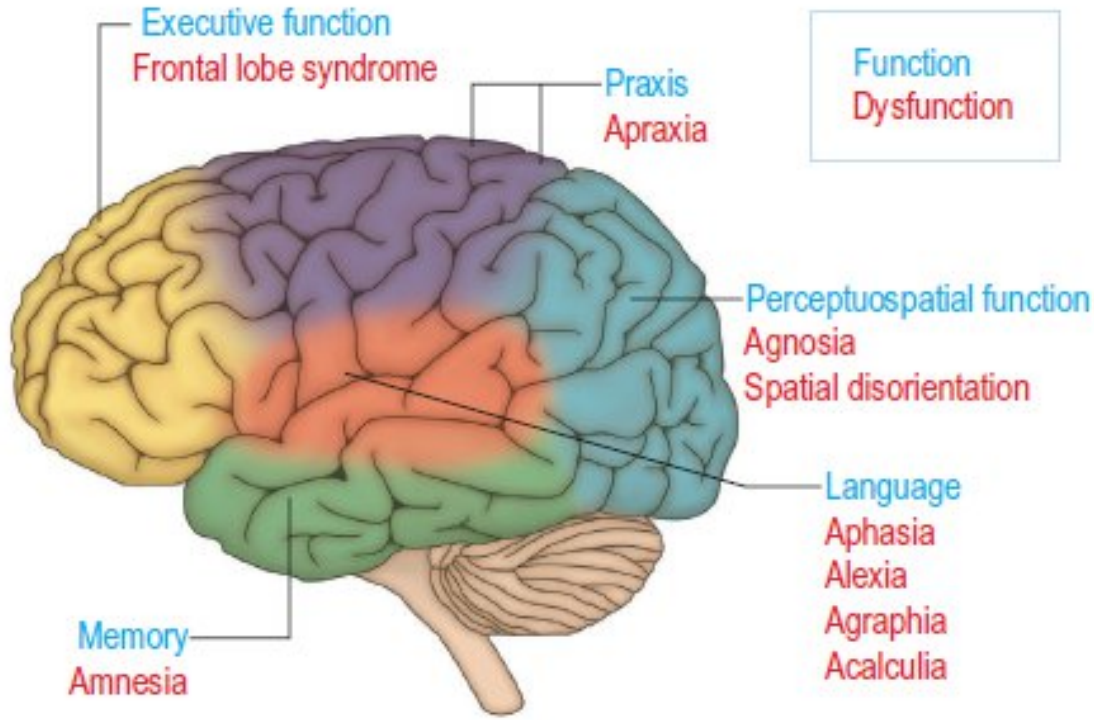


Figure 1.46 **Regional localisation of neuropsychological functions in the cerebral hemisphere and the syndromes associated with dysfunction.**